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Master Thesis

Creating an Individual Tree Crown Map for Leipzig Using YOLO26
Instance Segmentation and RGBI Aerial Orthophotos

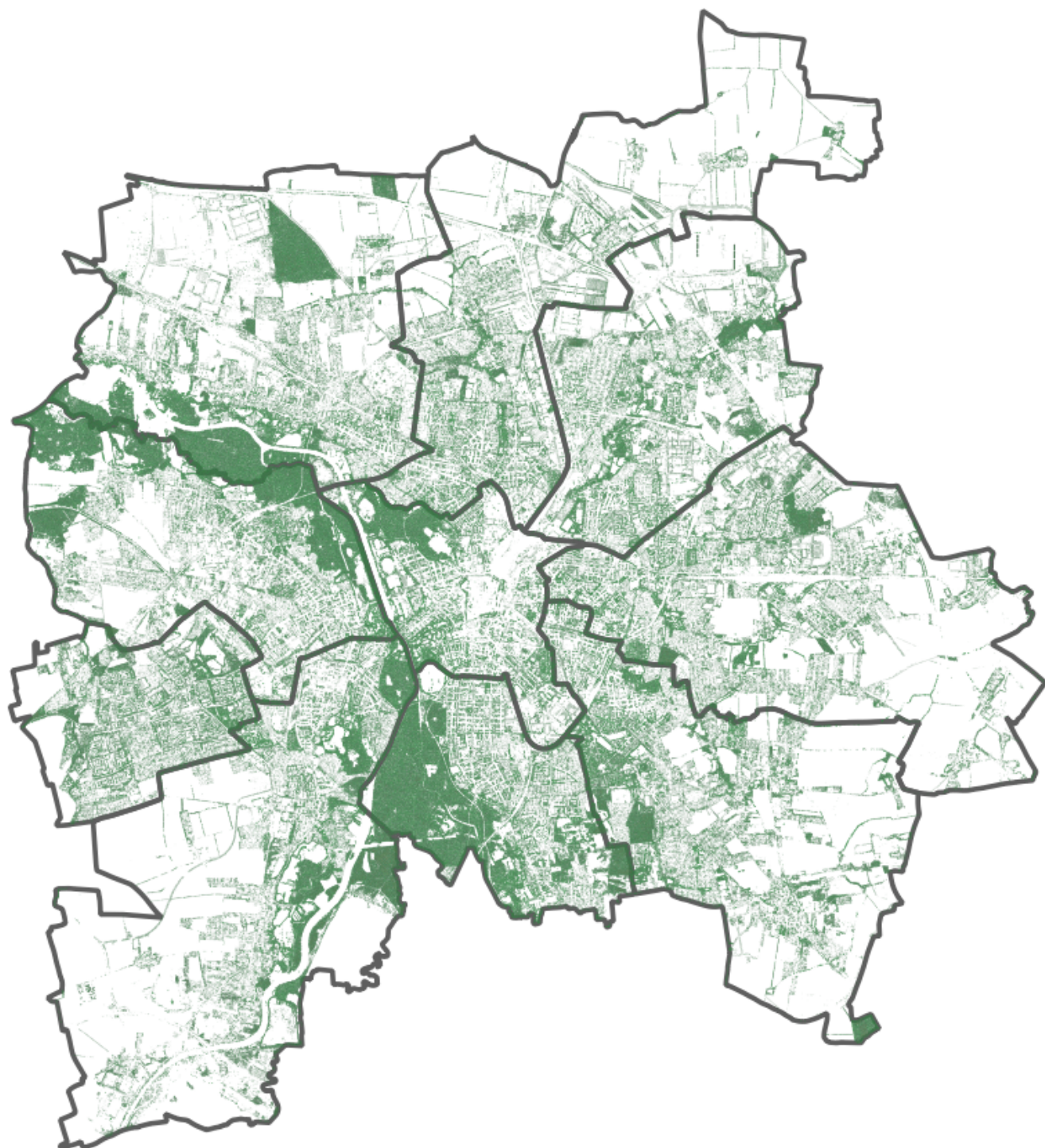
Abstract: This thesis developed and evaluated a YOLO26x-seg workflow for mapping individual tree crowns across Leipzig from red, green, blue, and near-infrared (RGBI) aerial orthophotos. The final workflow combined sequential hyperparameter tuning, multi-phase sliding-window inference, validation-supported confidence selection, and polygon post-processing. On the held-out test set, it achieved an F1 score of 0.66 and a mean matched intersection over union (IoU) of 0.75. The final Leipzig-wide map contained 951,265 mapped crowns with a summed crown area of 4,819.0 ha, corresponding to 16.2% of the city area.

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List of Abbreviations

AI	Artificial intelligence
ATKIS	Amtliches Topographisch-Kartographisches Informationssystem (Official topographic-cartographic information system)
Basis-DLM	Digitales Basis-Landschaftsmodell (Basic digital landscape model)
BGR	Blue, green, and red
COCO	Common Objects in Context
CRS	Coordinate reference system
CUDA	Compute Unified Device Architecture
DOI	Digital object identifier
DOP20	Digital orthophoto with a ground sampling distance of 20 cm
EPSG	European Petroleum Survey Group
ETRS89	European Terrestrial Reference System 1989
F1	F1 score
FN	False negative
FP	False positive
GeoTIFF	Georeferenced Tagged Image File Format
GPU	Graphics processing unit
GSD	Ground sampling distance
HPC	High-performance computing
HSV	Hue, saturation, and value
IoU	Intersection over union

IoU-NMS	Intersection-over-union-based non-maximum suppression
mAP	Mean average precision
mAP50	Mean average precision at an IoU threshold of 0.50
mAP50–95	Mean average precision averaged over IoU thresholds from 0.50 to 0.95
Mask R-CNN	Mask region-based convolutional neural network
MuSGD	Hybrid Muon–stochastic gradient descent optimiser
NDVI	Normalised difference vegetation index
NIR	Near-infrared
NMS	Non-maximum suppression
PCIe	Peripheral Component Interconnect Express
RGB	Red, green, and blue
RGBI	Red, green, blue, and near-infrared
TIFF	Tagged Image File Format
TP	True positive
UAV	Unmanned aerial vehicle
UFZ	Helmholtz Centre for Environmental Research
UTM	Universal Transverse Mercator
YOLO	You Only Look Once
YOLO26x-seg	Extra-large YOLO26 segmentation model

1 Introduction

This chapter introduces the scientific and methodological context of individual tree crown mapping in urban environments. It first outlines the need for spatially detailed information on urban trees and the potential of high-resolution aerial imagery to provide this information. The chapter then discusses deep-learning and You Only Look Once (YOLO)-based instance-segmentation approaches, followed by the challenges associated with applying these methods to large-area crown mapping. Finally, the research gap, study aim, research questions, and structure of the thesis are presented.

1.1 Urban trees and the need for individual-tree information

Urban trees are an important part of urban green infrastructure as they affect environmental conditions in densely built areas. Urban trees provide multiple ecosystem services. They can moderate urban temperatures and influence air-pollution dynamics, while urban forests also provide habitat and contribute to urban biodiversity (Alvey, 2006; Livesley et al., 2016).

However, urban trees are also exposed to increasing climatic stress, and climate change may increase their susceptibility to pests and diseases (Tubby & Webber, 2010). Urban vegetation is often fragmented and occurs within heterogeneous built environments, which complicates its assessment and management (Alegbeleye et al., 2026; Pirotti et al., 2026). Therefore, spatially detailed information is needed to identify changes and support informed decisions about the protection and management of urban tree resources (Alegbeleye et al., 2026; Stark et al., 2025).

Although conventional field surveys can provide detailed information on individual trees, they are time-consuming, labour-intensive and costly. This often limits their spatial coverage and complicates frequent updates across large urban areas (Alegbeleye et al., 2026; Pirotti et al., 2026; Zhao et al., 2023). Urban tree inventories also vary considerably in their scope and often focus on publicly managed trees, meaning that trees on private land may be underrepresented (Ma et al., 2021).

These limitations create a need for spatially continuous and regularly updatable approaches that can complement field-based inventories.

While canopy-cover maps are useful for showing the overall extent and distribution of urban vegetation, they do not contain the identity of individual trees. Semantic segmentation can distinguish between tree-covered and non-tree-covered areas, but neighbouring crowns of the same class are represented as a continuous canopy layer. Therefore, it cannot directly determine how many trees are present or where one crown ends and another begins (Troles et al., 2024). Individual tree crown delineation addresses this limitation by identifying each visible crown as a separate spatial object. This enables trees to be counted and their crown areas and dimensions to be measured individually. It also provides a spatial basis for linking further tree-level attributes (Dersch et al., 2023; Zhao et al., 2023). Consistently obtaining such information across an entire city requires methods that can identify and delineate individual crowns from remote sensing data.

1.2 Individual tree crown mapping from aerial imagery

High-resolution remote-sensing imagery provides a spatially continuous view of tree canopies and can therefore complement field-based inventories across larger areas. In contrast to ground surveys, aerial imagery captures the visible canopy over the entire study area in a consistent and georeferenced form. The high spatial resolution of modern imagery acquired from aircraft and unmanned aerial vehicles (UAVs) makes it possible to recognise crown shape, texture, colour, and the spatial relationships between neighbouring trees, providing the information required for individual tree crown mapping (Dersch et al., 2023; Liang et al., 2026).

Aerial imagery can be acquired from aircraft or UAV platforms, which differ in their spatial coverage and achievable resolution. UAVs can provide very detailed imagery at centimetre-level resolution and are therefore well suited to local or site-specific surveys, but their limited flight duration restricts the covered area. Aircraft-based imagery can cover larger areas in a consistent acquisition and is therefore well suited for city-wide mapping (Liang et al., 2026).

Visible and near-infrared bands provide complementary information for crown mapping. Red, green, and blue (RGB) imagery represents visual crown characteristics such as colour, brightness, texture, and shape, whereas near-infrared (NIR) reflectance provides additional information on vegetation that is not contained in the visible spectrum (Dersch et al., 2023; Liang et al., 2026). The inclusion of the NIR band provides vegetation-sensitive information and enables the calculation

of indices such as the normalised difference vegetation index (NDVI), which can support the assessment of canopy condition and vegetation activity (Yoon et al., 2026).

However, additional spectral information does not necessarily define the boundaries between neighbouring crowns where canopies overlap or appear similar (Gao et al., 2026; Liang et al., 2026).

The aim of individual tree crown mapping is to detect the presence of trees and assign each visible crown to a separate spatial object, delineating its boundary. This preserves the identity of individual trees and produces crown masks or polygons from which tree counts, crown areas, crown dimensions, and spatial distributions can be derived (Troles et al., 2024). These crown-level outputs can also provide a basis for linking information such as tree height, species, and vitality (Dersch et al., 2023), or vegetation indices (Gao et al., 2026), to individual trees.

Despite these advantages, it remains challenging to delineate individual crowns from optical imagery. Neighbouring crowns may touch or overlap, and similar colours and textures, shadows, and irregular crown shapes can obscure their boundaries. Smaller or partly occluded trees are particularly difficult to identify, especially in dense, structurally complex canopies (Do et al., 2026; Gao et al., 2026). Furthermore, optical imagery does not directly provide canopy height information, meaning crown separation relies mainly on spectral, textural and spatial differences visible in the image (Liang et al., 2026). These challenges have increased the need for methods that can account for complex patterns, motivating the growing use of deep learning for individual tree crown mapping.

1.3 Deep learning for individual tree crown detection and segmentation

Deep learning methods, particularly convolutional neural networks, have become increasingly popular for mapping individual tree crowns because they can learn the relevant features of images directly from annotated training data. These models learn to recognise hierarchical features such as crown colour, texture, shape, and spatial context. This enables them to account for a wider range of variation in tree appearance and environmental conditions than many traditional image-analysis techniques (Zhao et al., 2023).

Deep-learning approaches for tree crown mapping can be categorised as object detection, semantic segmentation or instance segmentation. Object-detection models can locate individual trees using bounding boxes, but they cannot delineate

the precise shape of the crown. Semantic segmentation models classify each pixel as either a tree or non-tree. However, they do not separate neighbouring crowns that belong to the same class. Instance segmentation combines these two tasks by identifying individual trees and assigning a separate pixel-level mask to each crown. This makes them particularly suitable for deriving crown-level spatial information (Troles et al., 2024; Zhao et al., 2023).

Instance-segmentation models can be broadly divided into two-stage and one-stage approaches. Two-stage methods, such as the Mask Region-Based Convolutional Neural Network (Mask R-CNN), first generate candidate regions, which are then classified and segmented. However, their repeated processing of individual regions increases model complexity and inference time. In contrast, one-stage approaches generate object detections and segmentation masks directly within a unified pipeline. Their more direct processing pipeline and faster inference speed make one-stage architectures attractive for individual tree crown mapping over large image datasets (Sun et al., 2025).

Despite these advances, the performance of deep-learning models depends strongly on the data used for training and evaluation. Differences in image resolution and quality, crown structure, tree density, illumination, and background conditions can all have a significant impact on segmentation accuracy. Furthermore, many existing studies have been conducted at only one or a few sites, meaning that the transferability of models to new environments remains uncertain. Benchmarking across diverse forest and urban conditions has revealed that generalisation to previously unseen imagery remains a significant challenge. However, local fine-tuning can considerably enhance performance when suitable annotated data are available (Burmeister et al., 2025; Liang et al., 2026; Wang et al., 2025).

A further limitation is the availability of sufficiently large and diverse annotated datasets. Because instance segmentation requires each tree crown to be delineated individually, creating reference data is highly demanding. The development of such datasets is becoming increasingly important for training and comparing instance-segmentation models across varied environments (Liang et al., 2026; Troles et al., 2024).

1.4 YOLO-based instance segmentation for tree crowns

The YOLO family comprises one-stage deep-learning models that predict objects directly within a unified network. The original YOLO framework formulated object detection as a single, end-to-end prediction task. This allows bounding boxes and class probabilities to be inferred directly from the entire image in a single pass through the network (Redmon et al., 2016). More recent versions also support instance segmentation, whereby a separate pixel-level mask is assigned to each detected object. This combination of object detection and mask prediction provides a favourable balance between segmentation performance and inference efficiency. This makes YOLO-based models well suited for processing high-resolution aerial imagery containing large numbers of tree crowns (Sun et al., 2025; Zhao et al., 2023).

Recent studies have adapted YOLO-based instance-segmentation models to the specific characteristics of tree crowns. These adaptations address variations in crown size and shape, overlapping canopies, and indistinct boundaries between neighbouring trees. For instance, YOLOv8E introduced network and candidate-selection adaptations to improve the segmentation of small and medium-sized crowns (Sun et al., 2025). DDB-YOLO incorporated specialised feature-extraction and crown-boundary components for dense and overlapping rubber-tree canopies (Gao et al., 2026). These studies show that YOLO-based models can efficiently delineate tree crowns.

Ultralytics introduced YOLO26 in 2026 as a new model generation with end-to-end inference without non-maximum suppression (NMS), improved assignment of small objects, and specific extensions for instance segmentation (Jocher et al., 2026). At the time of writing, no peer-reviewed study was identified that specifically evaluated YOLO26 for individual tree crown instance segmentation from aerial imagery. Its performance for this application therefore is largely unexplored.

1.5 Challenges of large-area individual tree crown mapping

Using an instance-segmentation model on a large orthomosaic requires more than just accurate predictions on individual image tiles. Since high-resolution aerial images are usually too large to be processed all at once, they are divided into smaller tiles, often with overlap. While this overlap preserves more context near tile edges, it can also result in the same crown being predicted multiple times or in fragmented and truncated masks being produced. Therefore, these tile-level predictions must be combined and corrected before they can form a consistent crown map. Combining tile-level predictions is not straightforward because duplicate predictions and neighbouring crowns may overlap in similar ways (He et al., 2026). Therefore, if duplicate-removal rules are too weak, the same crown may remain several times. If they are too strict, valid neighbouring crowns may be removed or merged. Post-processing must therefore remove duplicate predictions without losing separate trees.

The crown map also depends on the confidence threshold. A lower threshold retains more predictions but may also lead to more false detections. A higher threshold may remove valid crowns. Therefore, these settings should be selected using validation data rather than being applied as fixed default values (Wenkel et al., 2021).

As tiling and post-processing can affect the final predictions, the performance of the model on individual image tiles does not accurately reflect the quality of the resulting crown map. Therefore, the final outputs should also be evaluated after all predictions have been merged, using measures that reflect crown detection, boundary agreement, tree counts and mapped crown area (He et al., 2026; Liang et al., 2026).

1.6 Research gap and study motivation

Although deep-learning methods have produced promising results in individual tree crown mapping, many studies have focused on specific study areas, image types or canopy conditions. Models developed in these settings may therefore not be applicable to heterogeneous urban environments or imagery acquired using different sensors and resolutions (Liang et al., 2026; Wang et al., 2025). This restricts the application of results from individual experiments to city-wide crown mapping.

While existing YOLO-based studies have primarily focused on developing and evaluating models within specific datasets, fewer studies have examined the entire workflow required to consistently produce crown maps over large urban areas. In particular, the effects of tiled inference, duplicate removal, threshold selection and post-processing are often considered separately from model training and evaluation (Gao et al., 2026; He et al., 2026; Sun et al., 2025). This creates a need for an integrated workflow that connects model optimisation with large-area prediction and final map-level evaluation.

Leipzig provides a suitable case study because it combines heterogeneous urban areas with dense woodland.

1.7 Study aim and research questions

The aim of this study is to develop and evaluate a YOLO26-based workflow for mapping individual tree crowns across Leipzig from high-resolution red, green, blue, and near-infrared (RGBI) aerial orthophotos. The study covers the complete mapping process, including model training and optimisation, large-area sliding-window inference, polygon post-processing, evaluation of the final crown predictions, and production of a city-wide individual tree crown map.

The study addresses the following research questions:

1. To what extent does sequential hyperparameter tuning improve the mask-segmentation performance of YOLO26x-seg relative to the untuned baseline?
2. How accurately does the final prediction workflow delineate individual tree crowns across the complete evaluation dataset and within the Leipzig evaluation subset?
3. How does the performance of the final prediction workflow differ between urban/non-forest and forest evaluation contexts?
4. What Leipzig-wide crown totals, crown-size patterns, and differences between forest and urban/non-forest spatial contexts can be derived from the final crown map?

1.8 Thesis structure

The remainder of this thesis is organised into four chapters. Chapter 2 describes the study area, input data, data preparation, model selection, training and hyperparameter tuning, the large-area prediction workflow, and the evaluation methodology. Chapter 3 presents the effects of model optimisation, the performance of the final prediction workflow, the comparison between urban/non-forest and forest contexts, and the final Leipzig-wide individual tree crown map together with the derived crown patterns and summary statistics. Chapter 4 discusses the main findings, methodological limitations, and implications for large-area tree crown mapping. Chapter 5 summarises the conclusions and provides an outlook for future research.

2 Methods

This chapter describes the data and methods used to develop, evaluate, and apply the individual tree crown mapping workflow. It first introduces the Leipzig study area, the RGBI aerial orthophotos, and the tree crown reference data, before describing data preparation, model selection, training, and hyperparameter tuning. The subsequent sections present the large-area prediction and post-processing workflow, production of the Leipzig-wide crown map, the applied evaluation procedures, and the software and computational environment.

2.1 Study area

Leipzig, located in the Free State of Saxony in Germany, was selected as the target area for the final application of the developed individual tree crown mapping workflow. The municipal area extends approximately from 51.24° to 51.45° N and from 12.23° to 12.55° E.

The final prediction was carried out using the available RGBI aerial orthophoto coverage of the Leipzig city area.

As is typical of urban environments, Leipzig is characterised by a heterogeneous landscape. This includes individual trees, small tree groups, and larger, more continuous canopy areas, particularly in parks and wooded areas such as the Leipziger Auwald.

This variation in urban structure is important for individual tree crown delineation, because the difficulty of separating neighbouring crowns differs across the city. In open streets and green spaces, crowns are often spatially separated and their outlines are easier to recognise, while in dense vegetation, neighbouring crowns can merge visually, making the boundary between individual trees less clear.

Although Leipzig is the final mapping area of this study, the reference data used for model development and evaluation were not limited to Leipzig. They were collected from several locations in Germany to include a wider range of crown structures and image conditions. The following sections describe these areas and data.

2.2 Data

This study was based on RGBI aerial orthophotos and manually created reference data for individual tree crowns. The orthophoto tiles covered Leipzig as the final mapping area as well as the additional sites used to develop and evaluate the model.

The reference data consisted of manually delineated tree crown polygons. The polygons were created in QGIS by delineating visible individual tree crowns in the aerial imagery. Details on the orthophotos and the reference data are provided in the following sections. The subsequent preparation of images and labels for model training is described in Section [2.4](#).

2.2.1 RGBI aerial orthophotos

RGBI aerial orthophotos were used as image input for model development, evaluation, and the final prediction in Leipzig. The orthophoto tiles were downloaded as GeoTIFF files, a georeferenced form of the Tagged Image File Format (TIFF), from the official geospatial data portals of the respective German federal states. All imagery had a ground sampling distance (GSD) of 20 cm and contained four spectral bands: red, green, blue, and near-infrared (RGBI). The tiles were provided as 8-bit GeoTIFF files. The Leipzig imagery used for final map production consisted of digital orthophotos with a 20 cm ground sampling distance (DOP20) acquired in 2023. The Leipzig imagery was provided in the European Terrestrial Reference System 1989 / Universal Transverse Mercator (ETRS89 / UTM) zone 33N, corresponding to the European Petroleum Survey Group (EPSG) code 25833. The orthophotos from the additional reference-data sites covered different acquisition years and were provided either in ETRS89 / UTM zone 32N (EPSG:25832) or ETRS89 / UTM zone 33N (EPSG:25833), depending on the federal state and tile location.

Because individual tree crowns are strongly affected by phenology, the orthophotos were checked for suitability with respect to leaf-on conditions where possible. Imagery from the vegetation period was preferred, since leaf-off or early-spring imagery can reduce crown visibility and lead to incomplete detections. The acquisition years and coordinate reference systems (CRSs) of the orthophotos used in the study are summarised in Table 2.1.

Table 2.1 Acquisition years and coordinate reference systems of the RGBI aerial orthophotos used for the study sites.

Site or dataset	Acquisition year	CRS
Leipzig	2023	EPSG:25833
Auwald	2023	EPSG:25833
Chemnitz	2023	EPSG:25833
Hohes Holz	2017	EPSG:25832
Kalebsberg	2019	EPSG:25833
Rostock	2024	EPSG:25833
Tharandter Wald	2025	EPSG:25832 EPSG:25833

2.2.2 Tree crown reference data

The tree crown reference data were created manually in QGIS by visually delineating individual tree crowns in the RGBI aerial orthophotos. Each crown was stored as a polygon representing its visible extent in the imagery. These manually delineated polygons served as the reference data for model development and evaluation.

Reference data were available for seven sites in Germany: Auwald, Chemnitz, Hohes Holz, Kalebsberg, Leipzig, Rostock, and Tharandter Wald. The final prepared dataset comprised 397 labelled image tiles containing 22,864 individual tree crown polygons (Table 2.2). All reference features were polygon geometries. The reference data covered different vegetation structures, ranging from more open tree cover to dense and connected canopies.

Table 2.2 Number of labelled image tiles and manually delineated tree crown polygons available for each reference-data site.

Site	Labelled image tiles	Tree crown polygons
Leipzig	275	9,878
Auwald	14	1,969
Chemnitz	17	998
Hohes Holz	17	2,235
Kalebsberg	26	3,343
Rostock	36	2,002
Tharandter Wald	12	2,439
Total	397	22,864

2.3 Overall study workflow

The overall workflow of this study is summarised in Figure 2.1. It was organised into three main parts: data preparation, model training, and the final inference pipeline.

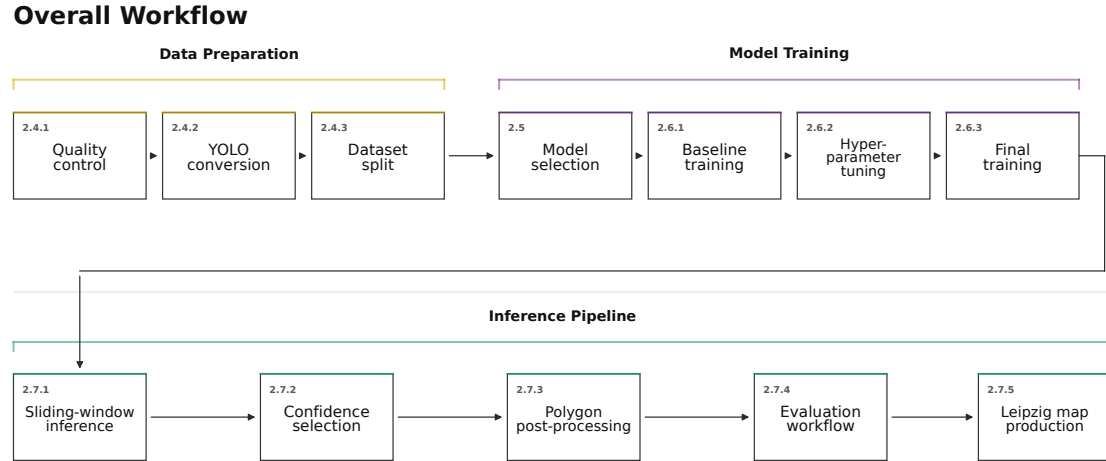


Figure 2.1 Overall workflow of the study, comprising data preparation, model training, and the final inference pipeline.

The data preparation workflow included quality control and standardisation of the input data, conversion of the imagery and tree-crown polygons into the YOLO image and label formats, and the creation of training, validation, and test datasets. These steps are described in Section 2.4.

Model development started with the selection of the YOLO26x-seg model architecture. A baseline model was trained first, followed by hyperparameter tuning using the training and validation data. The selected configuration was then used to train the final YOLO26x-seg model. Model selection is described in Section 2.5, while baseline training, hyperparameter tuning, and final model training are described in Section 2.6.

After model training, the final flow was applied to larger orthophoto areas. This inference pipeline included multi-phase sliding-window prediction, confidence-threshold selection, polygon post-processing, evaluation of the final flow, and production of the final Leipzig crown map. These steps are described in Section 2.7.

2.4 Data preparation

This section describes how the aerial orthophotos and tree crown reference data were prepared for use with the YOLO26 instance-segmentation model. This included quality control and standardisation of the source data, conversion of the image and annotation files into the required YOLO format, and creation of the training, validation and test datasets.

2.4.1 Quality control and standardisation

Before preparing the data for model training, the aerial orthophotos and their associated tree crown labels were verified for completeness, spatial accuracy, and consistency. Each labelled image tile was matched with its corresponding annotation file, and the alignment between the imagery and crown polygons was visually verified using QGIS.

The image tiles were standardised to a size of 500×500 pixels. The associated crown annotations were prepared to correspond with the final image extent, ensuring that the image and label boundaries were aligned.

The reference polygons were then checked to ensure that they had valid geometry and correctly represented individual tree crowns. This step ensured that the image tiles and their annotations were ready for conversion to the YOLO segmentation format.

2.4.2 Conversion to YOLO image and label format

After quality control, the RGBI aerial orthophotos and their associated crown labels were converted into the required YOLO26 instance segmentation format. The final dataset was stored as four-channel GeoTIFF image files. The original band order of red, green, blue, and near-infrared (RGBI) was retained. YOLO models are commonly applied to three-channel RGB imagery. However, the Ultralytics implementation used in this study supported multi-channel image inputs directly. Therefore, RGBI imagery could be used as a four-channel input without requiring manual changes to the YOLO26 model code. The adaptation of the first convolutional layer for four input channels is described in Section 2.6.1. The manually delineated crown polygons were converted from GeoJSON files into YOLO segmentation labels. Pixel-coordinate encoding was used for label conversion. Each polygon vertex was converted from its map position to a row and column position

within the corresponding image tile. These pixel coordinates were then normalised by the image width and height to produce values between 0 and 1.

Each polygon was assigned to the “tree crown” class and stored as a single YOLO segmentation label record. Image and label files shared identical base file names. This ensured a clear correspondence between each four-channel image tile and its respective segmentation label file.

2.4.3 Creation of training, validation, and test datasets

The prepared dataset was divided into training, validation and test subsets. Each image tile and all associated tree crown annotations were assigned to one subset.

The final split was approximately 70/15/15 at the image-tile level. This design is comparable to that used in recent urban tree detection work (Stark et al., 2025), where 70% of the data were also used for training, 15% for validation, and 15% for testing. In this study, the resulting proportions were 70.4% for training, 14.9% for validation and 14.7% for testing.

The split was area-stratified. First, image tiles were grouped by study area. Then, the training, validation, and test subsets were created so that each subset contained a comparable representation of the different study areas. This reduced the risk of obtaining a favourable or unfavourable split by chance. It also ensured that the results of the validation and testing phases were not dominated by a single study area or crown condition.

Alongside the labelled image tiles, the training subset included five negative background-only image tiles without tree crown annotations. These tiles were stored with empty YOLO label files and represented image areas without labelled tree crowns. These tiles accounted for 1.2% of all image tiles and 1.8% of the training subset. The background-only tiles were only included in the training subset and were not present in the validation or test subsets.

The final dataset contained 402 image tiles, of which 397 contained crown annotations and five were background-only training tiles. In total, the dataset included 22,864 tree crown annotations. The training subset comprised 283 image tiles and 16,309 crown annotations. The validation subset comprised 60 image tiles and 3,038 crown annotations, while the held-out test subset comprised 59 image tiles and 3,517 crown annotations (Table 2.3).

Table 2.3 Final dataset composition by subset.

Subset	Image tiles	Share (%)	Tiles with crowns	Background tiles	Crown annotations
Training	283	70.4	278	5	16,309
Validation	60	14.9	60	0	3,038
Test	59	14.7	59	0	3,517
Total	402	100.0	397	5	22,864

2.5 Selection of the YOLO26x-seg model

The YOLO26 model is available in several scales, ranging from nano (n) to extra large (x) (Jocher et al., 2026). The Ultralytics implementation provides these scales for instance segmentation models too, including the YOLO26x-seg variant. Larger model variants generally have a higher representational capacity, but require more computational resources and longer training and inference times.

Using a large YOLO variant is also consistent with recent work on individual tree crown segmentation, where high-capacity YOLO models have been applied to improve crown-delineation performance (Sun et al., 2026). This demonstrates the relevance of high-capacity YOLO variants for tree crown segmentation tasks.

Preliminary training experiments were conducted with different YOLO26 segmentation model scales using the prepared tree crown dataset. The YOLO26x-seg variant achieved the strongest validation performance across these experiments. This variant was therefore selected as the model scale to be used throughout the main model-development workflow and for the final Leipzig-wide prediction.

Selecting YOLO26x-seg reflected this study’s aim to prioritise crown delineation performance over minimum training and inference time. The necessary computational resources were provided by the high-performance computing environment outlined in Section 2.9.

2.6 Model training and hyperparameter tuning

This section describes the training and optimisation of the selected YOLO26x-seg model using the prepared four-band RGBI dataset. The model was developed in three stages: training a controlled baseline model; systematic hyperparameter tuning; and final model training using the selected configuration.

All training runs used the same area-stratified training, validation, and test datasets described in Section 2.4.3. The fixed training setup comprised the YOLO26x-seg model architecture; four-band RGBI input; pretrained model initialisation; an input image size of 1024 pixels; a batch size of 16; a maximum training duration of 200 epochs; the hybrid Muon–Stochastic Gradient Descent (MuSGD) optimiser; and a fixed random seed to ensure reproducibility. Validation data were used during training to monitor model performance and select the best-performing checkpoint. The held-out test dataset was not used during model training or hyperparameter tuning.

For each training run, the epoch with the highest validation fitness was identified and the corresponding checkpoint was retained as the best model checkpoint. In the Ultralytics implementation used, validation fitness was calculated as described in Section 2.8.2. Thus, checkpoint selection considered both bounding-box localisation and instance-mask delineation performance.

The following subsections describe the controlled baseline model, the hyperparameter-tuning procedure, and the final model training using the selected parameter configuration.

2.6.1 Baseline model training

A controlled baseline model was trained before hyperparameter tuning to provide a reference for assessing the effect of the optimisation procedure. The baseline used the same model architecture, RGBI input data, dataset split, and fixed training configuration described in Section 2.6. Hyperparameters that were subsequently included in the tuning procedure were left at their Ultralytics default values. The resulting model therefore represents a controlled baseline.

The model was initialised from the pretrained yolo26x-seg.pt checkpoint. Ultralytics YOLO26 segmentation checkpoints use the `-seg` suffix and are provided as models pretrained on the Common Objects in Context (COCO) dataset before being adapted to user-specific segmentation tasks (Jocher et al., 2026). Although this pretraining was not specific to aerial tree crowns or RGBI imagery, it provided an established starting point for fine-tuning. The use of COCO-based initiali-

sation or COCO-derived YOLO training data has also been reported in recent vegetation-related detection and segmentation studies, including orchard fruit detection (Dinataputra et al., 2026) but also aerial forest crown-damage mapping (Naseri et al., 2026).

To accommodate the four-band RGBI input, the first convolutional layer was modified to have four input channels instead of three. The pretrained weights associated with the red, green and blue channels were retained, while the newly added near-infrared channel was initialised and learned during training on the tree crown dataset. The model was then trained to predict the single target class of individual tree crowns.

2.6.2 Hyperparameter tuning

Hyperparameter tuning was conducted to improve performance over the described baseline model. The tuning space included a large number of training, loss-function, and augmentation hyperparameters. Testing all possible combinations would have required an impractically large number of complete model-training runs. A sequential one-factor-at-a-time tuning strategy was therefore used.

The initial tuning configuration was based on informed parameter selection and results from previous exploratory model-development experiments. It served as the starting point for the sequential tuning procedure.

During this procedure, one hyperparameter was varied while all other settings were held fixed. For the first parameter, the remaining settings corresponded to the initial tuning configuration. After a value had been selected, it was added to the current configuration and retained while the next hyperparameter was examined. Each tuning stage therefore started from the best configuration identified in the preceding stage.

Tuning began with the initial learning rate (1r0). For each hyperparameter, six candidate values were first evaluated over a comparatively coarse range. Based on validation performance, four additional values were then tested around the strongest coarse candidate. Each hyperparameter was therefore evaluated using ten candidate values in total. The complete validation-fitness patterns and selected values from the sequential tuning procedure are shown in Figure A.1.

Each candidate was the result of an independent, complete training run. The models were always initialised from the pre-trained yolo26x-seg.pt checkpoint and no model weights, optimiser states or scheduler states were transferred between candidates. All candidates used the same training and validation data, model architecture, image size, batch size, optimiser, and random seed as reported in

Section 2.6. The maximum training duration was also kept fixed, except for the final tuning stage in which the maximum number of epochs was evaluated.

The adaptive one-factor-at-a-time approach allowed a large set of hyperparameters to be investigated within the available computational budget (Frey et al., 2003). However, because the parameters were not evaluated jointly, the procedure did not systematically account for interactions among them. The selected configuration should therefore be regarded as a practically optimised parameter set rather than a demonstrated global optimum.

2.6.3 Final model training

Once the sequential hyperparameter tuning was complete, the selected parameter values were combined to create the final training configuration. A final YOLO26x-seg model was then trained using this configuration and the fixed training setup described in Section 2.6.

Rather than from one of the tuning candidates, the final model was initialised independently from the pretrained yolo26x-seg.pt checkpoint. The selected hyperparameters included the values identified during the tuning procedure described above, while all fixed settings, including the model architecture, input configuration, image size, batch size, maximum number of epochs, optimiser and random seed, remained unchanged.

Validation data were used during the final training stage to monitor performance and identify the epoch with the highest validation fitness. The corresponding best.pt checkpoint was retained as the final model. The held-out test dataset was not used during final model training or checkpoint selection.

The selected hyperparameter configuration corresponds to the best-performing values shown in Appendix Figure A.1. This model was then used as the fixed input for the subsequent flow, in which large-area inference, confidence-threshold selection, polygon post-processing, evaluation prediction, and final crown-map production were performed.

2.7 Prediction workflow and map production

The final trained YOLO26x-seg model was used for a custom flow to generate individual tree crown polygons from large RGBI orthophotos. The workflow was designed to apply the model beyond the prepared 500×500 pixel training tiles and to produce spatially coherent crown predictions over larger image areas.

The prediction pipeline consisted of five main components. First, large-area inference was performed using overlapping image windows and two shifted prediction phases. Second, a validation-based confidence-threshold analysis was used to select the final confidence threshold. Third, geometric post-processing was applied to reduce duplicate and overlapping crown polygons. Fourth, validation and test predictions were generated using the same large-area workflow to evaluate the final prediction pipeline under conditions similar to final map production. Finally, the Leipzig crown map was produced, which enabled a city-wide crown summary analysis.

The following sections describe these components in the order in which they were applied.

2.7.1 Multi-phase sliding-window inference

The large-area inference workflow was used for both evaluation prediction and final map production. Input imagery consisted of DOP20 RGBI orthophotos obtained from the official geodata portals of the respective federal states. For prediction, the YOLO inference image-size parameter was set to 2048 pixels. This preserved the scale factor of approximately 2.05 between the original image dimensions and the network input that was used during training, because the 500-pixel training tiles were resized to 1024 pixels and the 1000-pixel inference windows were resized to 2048 pixels. Crowns therefore appeared at approximately the same scale to the model during training and inference.

Each source image was divided into overlapping square windows of 1000×1000 pixels. At the 20 cm spatial resolution of the DOP20 imagery, this corresponds to 200×200 m. Adjacent windows overlapped by 500 pixels, corresponding to 100 m. This overlap ensured that crowns located close to the edge of one window were also observed in the neighbouring window.

Prediction was performed in two phases (Figure 2.2). The first phase used a regular window grid. The second phase used the same window size and overlap but shifted the grid by 250 pixels, corresponding to 50 m. This shifted phase increased the probability that crowns located close to a window edge in one phase were located closer to the centre of a window in the other phase.

Multi-phase Sliding-Window Inference



Window size: 1000 px = 200 m; overlap: 500 px = 100 m; phase shift: 250 px = 50 m.

Figure 2.2 Multi-phase sliding-window inference. Overlapping 1000×1000-pixel windows were processed with an overlap of 500 pixels. In the second prediction phase, the window grid was shifted by 250 pixels in both spatial directions.

For each prediction window, image bands were read in red, green, blue, and near-infrared order and passed to the trained YOLO26x-seg model. Predicted masks touching an internal window edge were removed when the edge contact exceeded 15 pixels. This edge filter eliminated incomplete crown predictions caused by window-border truncation. The second shifted prediction phase compensated for this removal, because crowns removed near a window boundary in one phase could still be predicted from a more central position in the other phase. After prediction, mask coordinates were transformed back into georeferenced polygon geometries. Predictions from all windows and both phases were collected into one candidate set before post-processing.

2.7.2 Confidence-threshold selection

The confidence threshold controls which crown candidates are retained before post-processing. Lower thresholds retain more possible crowns but also more low-confidence predictions, while higher thresholds remove uncertain predictions but may also discard true crowns. A validation-supported threshold analysis was therefore conducted using the multi-phase inference workflow described in Section 2.7.1. Thresholds were tested with fixed post-processing settings: global polygon intersection-over-union-based non-maximum suppression (IoU-NMS) = 0.50 and minimum retained-area fraction after overlap subtraction = 0.70. A detailed description of those two post-processing settings is reported in Section 2.7.3. The resulting post-processed validation predictions were summarised using an F1-confidence curve (Figure 2.3).

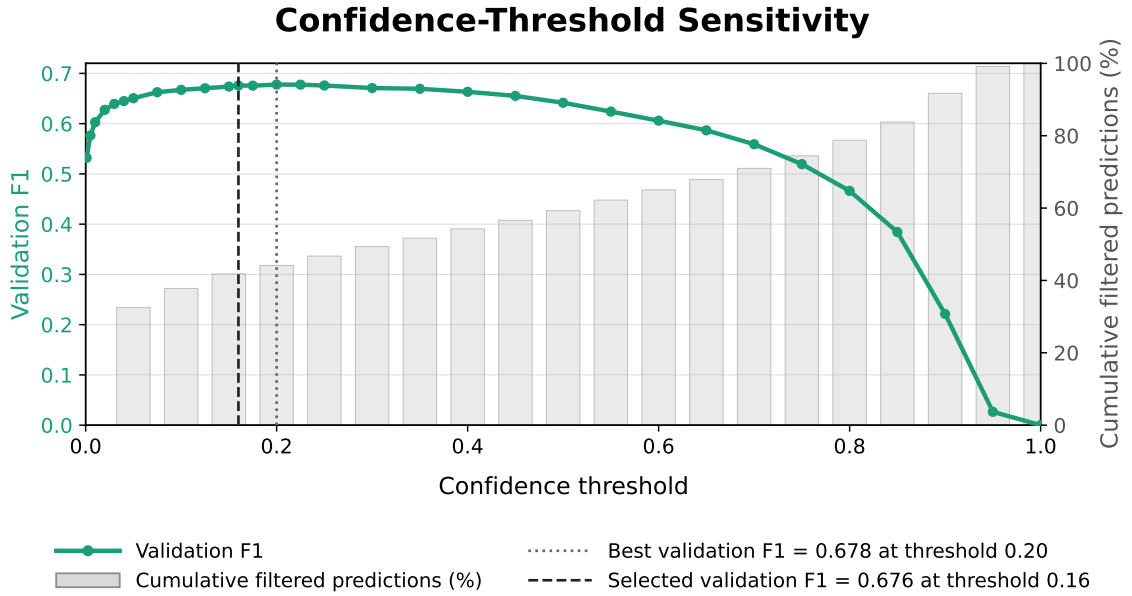


Figure 2.3 Post-processed validation F1 across confidence thresholds, generated with multi-phase sliding-window inference and fixed polygon post-processing settings: IoU-NMS = 0.50 and minimum retained-area fraction = 0.70. The cumulative share of filtered predictions is shown alongside the F1 curve.

The highest validation F1 score was reached at a confidence threshold of 0.20. However, the curve showed only a very small difference between this maximum and the value obtained at a threshold of 0.16. The final confidence threshold was therefore set to 0.16. This value was selected because it retained a slightly more recall-oriented prediction set while remaining close to the best validation F1 value.

Precision values were interpreted carefully during this selection. The reference labels are not a perfect representation of all visually detectable crowns. Some predictions counted as false positives may therefore represent plausible tree crowns that were missing from the reference labels or delineated differently during reference creation. For this reason, the final confidence threshold was selected based on both validation metrics and visual inspection rather than by the numerical F1 maximum alone.

2.7.3 Polygon post-processing

Post-processing was applied after multi-phase inference to eliminate low-confidence, duplicate and overlapping crown predictions. First, predictions from all windows and both phases were combined into one georeferenced candidate set. The fixed confidence threshold selected in Section 2.7.2 was then applied, followed by geometric post-processing.

The first geometric post-processing step was global, polygon-based non-maximum suppression. This is a common step in object-detection workflows, where it is used to reduce multiple detections of the same object by retaining the highest-confidence candidate and suppressing lower-confidence candidates that overlap it beyond a specified threshold (Bodla et al., 2017). In this workflow, suppression was applied to crown polygons rather than bounding boxes. Pairwise overlap was measured using polygon intersection over union, and lower-confidence polygons were removed when they overlapped an already retained polygon by more than the global IoU-NMS threshold.

The second geometric post-processing step was overlap subtraction. The remaining predictions were processed in descending order of confidence. Higher-confidence polygons were accepted first. For each subsequent polygon, areas that had already been covered by the higher-confidence polygons that had been accepted were subtracted from its geometry. The polygon was retained only if at least a certain proportion of its original area remained after subtraction. This parameter is referred to as the minimum retained-area fraction after overlap subtraction.

The retained-area fraction was then used to remove small leftover fragments caused by duplicate predictions. At the same time, it allowed plausible, non-overlapping parts of neighbouring crown predictions to remain instead of removing every polygon that partially overlapped a higher-confidence prediction. The global polygon IoU-NMS threshold and minimum retained-area fraction were selected based on a combination of quantitative evaluation on the validation set and visual examination of the resulting crown polygons. This ensured that the selected settings provided suitable validation performance while also producing spatially plausible crown boundaries and effectively reducing duplicate predictions. The final post-processing configuration used for both evaluation and Leipzig-wide crown-map production is summarised in Table 2.4.

Table 2.4 Final confidence-threshold and polygon post-processing configuration used for evaluation and Leipzig-wide crown map production.

Parameter	Selected value
Confidence threshold	0.16
Global polygon IoU-NMS threshold	0.50
Minimum retained-area fraction after overlap subtraction	0.70

These settings were applied consistently throughout the evaluation workflow and during production of the final Leipzig crown map.

2.7.4 Generation of validation and test predictions

The evaluation workflow was designed to assess the final prediction pipeline under conditions similar to those of the final large-area map production process. Therefore, the validation and test labels were not evaluated only as individual 500×500 pixel image tiles. Instead, the reference tiles were linked back to their larger source orthophoto areas, and larger source crops were prepared around the locations of the validation and test tiles.

The multi-phase inference workflow described in Section 2.7.1 was then applied to the larger source crops. This ensured that the evaluation predictions were generated using the same logic for overlapping windows, edge filtering, prediction aggregation, confidence thresholding, and post-processing as in the final Leipzig map workflow. This approach accounted for effects arising during large-area prediction, such as duplicate detections from overlapping windows and incomplete predictions close to window borders.

Following prediction and post-processing, the resulting crown polygons were clipped back to the original extents of the validation and test tiles. Performance could therefore be calculated only within the held-out reference tile areas, while the predictions were generated using a larger surrounding image context. This made the evaluation more comparable to the final mapping workflow than directly predicting isolated reference tiles.

Evaluation predictions were prepared for two main spatial scopes. The first scope comprised the complete evaluation set, including all available validation and test tiles from all study areas. This scope was used to assess the general performance of the final workflow across the image and crown-structure conditions represented in the dataset. The second scope comprised the Leipzig map-area evaluation and included only the validation and test tiles located inside the final Leipzig mapping area. This scope was used to estimate the performance most directly relevant to the final Leipzig crown map. The results for both spatial scopes are presented in Section 3.2.

In addition, the held-out test set was divided into urban/non-forest and forest site contexts to examine whether prediction performance differed between the two broader environmental settings. The urban/non-forest context comprised the test tiles from Leipzig, Chemnitz, and Rostock, while the forest context comprised the test tiles from Auwald, Hohes Holz, Kalebsberg, and Tharandter Wald. The results of the site-context comparison are presented in Section 3.3.

The resulting prediction outputs were evaluated using the crown-level matching rules and pooled metrics described in Section 2.8.3.

2.7.5 Final Leipzig-wide crown map production

The final Leipzig crown map was produced separately for each of the ten Leipzig districts. Each district was prepared individually. This kept the large-area prediction workflow computationally manageable and enabled district-specific map production.

Before prediction, a 250 m buffer was added around each district. This ensured that tree crowns close to administrative boundaries could still be predicted with the appropriate surrounding image context. Without this buffer, crowns located near a district edge could have been affected by artificial crop boundaries during the prediction process.

The buffered district images were processed using the multi-phase inference workflow and fixed post-processing settings described in Sections 2.7.1 and 2.7.3, respectively. Following prediction and post-processing, the crown polygons were

assigned to the relevant Leipzig district based on their centroid location. This centroid-based ownership rule ensured that each crown was assigned to one district without splitting crowns at district boundaries.

For the final map output, the district-level prediction layers were restricted to the Leipzig city extent. The final Leipzig crown map therefore consists of the centroid-assigned crown polygon outputs for all ten districts. The resulting map is shown in Section 3.4. The vector dataset was exported as a GeoPackage and is publicly available on Zenodo. The corresponding DOI is provided in the [Data and code availability section](#).

2.7.6 Leipzig-wide crown summary and spatial-context analysis

The final post-processed crown polygons were used to derive summary statistics for the complete Leipzig mapping area. Crown size was represented by the area of each mapped crown polygon in square metres. Crown count, summed crown area, crown cover, mean and median crown size, and the crown-size distribution were calculated. Crown cover was defined as the summed mapped crown area divided by the corresponding spatial area and expressed as a percentage.

The crown-size distribution was visualised using 1 m² bins normalised to the percentage of mapped crowns per bin. Crown sizes larger than 250 m² were excluded from the histograms to improve readability but remained included in all summary statistics. Local modes in the Leipzig-wide distribution were identified directly from the histogram bins.

For an additional spatial comparison, the Leipzig mapping area was divided into forest and urban/non-forest areas. Forest areas were derived from the official *Amtliches Topographisch-Kartographisches Informationssystem* (ATKIS) Digitales Basis-Landschaftsmodell (Basis-DLM) forest polygons (*AX_Wald*, shapefile layer *VEG02_F*), downloaded from the GeoSN Basis-DLM data portal. The forest polygons were clipped to the municipal boundary of Leipzig in QGIS, while the remaining municipal area was classified as urban/non-forest. Each mapped crown was assigned to one of the two spatial contexts according to the location of its polygon centroid. Crown count, summed crown area, crown cover, mean and median crown size, and the crown-size distribution were calculated separately for both spatial contexts. Crown cover was calculated relative to the total area of the respective spatial context.

The Leipzig-wide results are presented in Section 3.4.1, while the spatial-context comparison is presented in Section 3.4.2.

2.8 Performance evaluation

The evaluation was divided into two parts because different stages of the workflow required different metrics. Standard YOLO mean average precision (mAP) metrics, including mAP50 and mAP50–95, were used to evaluate and compare model training results, as described in Section 2.8.2. These metrics assessed model performance before the complete prediction and post-processing workflow was applied. The final post-processed crown polygons were therefore evaluated separately using custom crown-level and map-level metrics that better describe the quality of the final inference outputs, as presented in Section 2.8.3. Section 2.8.1 first describes the separation between training, validation, and test data and explains how these subsets were used during model development and final evaluation.

2.8.1 Evaluation design and data separation

As introduced in Section 2.4.3, the dataset was divided into training, validation, and test subsets prior to model training. The training subset was used to fit the model weights. The validation subset was used for model development, including checkpoint selection, tuning the hyperparameters, and calibrating the post-processing. The test subset, which was held out, was reserved for final evaluation after the model configuration and flow had been finalised. This separation was employed to prevent the selection of model or post-processing settings based on test performance. The test results were therefore used to evaluate the performance of the final fixed workflow on unseen reference tiles.

2.8.2 Training evaluation metrics

The Ultralytics segmentation framework’s standard validation outputs were used to evaluate model training and hyperparameter tuning. During training, the model was evaluated on the validation subset after each epoch, with the checkpoint showing the highest validation fitness being saved as the best model. For segmentation models, the Ultralytics metric output includes both box- and mask-based metrics. Mask metrics are reported as mask precision, mask recall, mask mAP50 and mask mAP50–95. In Ultralytics output, these are denoted by the suffix ‘M’, while box metrics are denoted by the suffix ‘B’. The segmentation fitness score combines the mask fitness score with the corresponding box fitness score. In the version used in this study, the resulting fitness value was:

$$\text{fitness} = \text{mAP}_{50-95}^{\text{box}} + \text{mAP}_{50-95}^{\text{mask}}.$$

This validation fitness was used for checkpoint selection and comparison between tuning configurations.

As the task involved individual tree crown segmentation, the interpretation of training performance focused on mask-based metrics. Mask precision describes the proportion of predicted mask instances that were correct according to the Ultralytics validation procedure. Low mask precision indicates that a model produced many false-positive mask predictions. Mask recall describes the proportion of reference instances detected by the model. Low recall indicates that the model missed many reference crowns.

In addition to precision and recall, the mask F1 score was used as a combined measure of detection performance. F1 is the harmonic mean of precision and recall and is high only when both metrics are high. It was therefore useful for comparing models with different balances between false-positive and missed crown predictions.

Mask mAP summarises performance over the precision–recall curve. Mask mAP50 is calculated using an IoU threshold of 0.50, meaning that a predicted mask is considered sufficiently similar to a reference instance at this threshold. Mask mAP50–95 is a stricter metric because it averages mAP over ten IoU thresholds ranging from 0.50 to 0.95 in 0.05 increments. This metric therefore rewards not only detecting the correct instances, but also more precise mask localisation.

The training-evaluation metrics reported in this study were mask precision, mask recall, mask F1, mask mAP50, and mask mAP50–95. These metrics were used to compare baseline training and the final tuned model. As they describe model performance before the final large-area inference and post-processing workflow was applied, they were treated separately from the final workflow evaluation described in Section 2.8.3.

2.8.3 Evaluation of the final prediction workflow

The final workflow evaluation assessed the georeferenced crown polygons produced by the complete prediction workflow described in Section 2.7. In contrast to the training evaluation, this assessment was applied after multi-phase inference, confidence thresholding, and geometric post-processing. It therefore evaluated the final crown-polygon outputs rather than the raw model performance.

The inference evaluation was based on object-level comparison between predicted crown polygons and reference crown polygons. Therefore, a matching procedure was required to determine which predicted crowns corresponded to which reference crowns.

2.8.3.1 Crown-level IoU matching

Predicted and reference crown polygons were matched using polygon intersection over union (IoU). For each prediction-reference pair, IoU was calculated as the area of overlap divided by the area covered by the union of both polygons. A prediction was considered eligible for matching with a reference crown when their polygon IoU was at least 0.50. This threshold follows the common convention in object-detection-based tree crown delineation studies, where an IoU threshold of 0.50 is frequently used to define matches. Using this threshold also supports comparability with previous studies that evaluated individual tree crown predictions using the same matching criterion (Ball et al., 2023; Braga et al., 2020; Chi et al., 2024; Weinstein et al., 2020).

Matching was performed separately within each reference tile. All eligible prediction-reference pairs were sorted in descending order by IoU. Pairs were then accepted one by one, starting with the highest IoU value. A pair was accepted only if neither the predicted polygon nor the reference polygon had already been matched. This produced a one-to-one assignment between predicted and reference crowns.

Accepted prediction-reference pairs were counted as true positives (TP). Predicted crowns that were not matched to any reference crown were counted as false positives (FP). Reference crowns without a matched prediction were counted as false negatives (FN). The resulting TP, FP and FN counts are the basis for the inference evaluation metrics described in the following section.

2.8.3.2 Final workflow evaluation metrics

The main inference metrics were calculated from the matched and unmatched crown polygons described in Section 2.8.3.1. For each evaluated dataset subset, true positives, false positives, false negatives, predicted crown counts, reference crown counts, predicted crown area, and reference crown area were first summed across all included tiles. The final metrics were then calculated from these summed values.

Precision was calculated as the proportion of predicted crowns that were matched to a reference crown:

$$P = \frac{TP}{TP + FP}.$$

Recall was calculated as the proportion of reference crowns that were detected by a matching prediction:

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

The F1 score was used to summarise the balance between precision and recall:

$$F_1 = \frac{2PR}{P + R}.$$

In addition to the detection metrics, the mean matched IoU was calculated for all accepted prediction–reference pairs:

$$\overline{\text{IoU}}_{\text{matched}} = \frac{1}{\text{TP}} \sum_{i=1}^{\text{TP}} \text{IoU}_i.$$

Here, IoU_i denotes the Intersection over Union of the i -th accepted prediction–reference match. The mean matched IoU describes the average geometric overlap of successfully matched crowns and therefore provides information on crown-shape agreement after matching.

Crown-count agreement was evaluated using the crown-count ratio:

$$\text{Count ratio} = \frac{N_{\text{pred}}}{N_{\text{ref}}}.$$

Here, N_{pred} is the total number of predicted crowns and N_{ref} is the total number of reference crowns in the evaluated subset. A value below 1 indicates underprediction of crown counts, whereas a value above 1 indicates overprediction.

Crown-area agreement was evaluated using the crown-area ratio:

$$\text{Area ratio} = \frac{A_{\text{pred}}}{A_{\text{ref}}}.$$

Here, A_{pred} is the summed predicted crown area and A_{ref} is the summed reference crown area. A value below 1 indicates underestimation of total crown area, whereas a value above 1 indicates overestimation.

These pooled metrics were used as the main quantitative summary of inference performance. They were calculated from the total number of crowns and total

crown area in each evaluated subset rather than as averages of individual tile-level metrics.

2.9 Software and computational environment

All data preparation, model training, prediction, and evaluation workflows were implemented in Python 3.12.13. YOLO26 instance-segmentation models were trained and applied using the Ultralytics framework version 8.4.47 with PyTorch 2.11.0 compiled for Compute Unified Device Architecture (CUDA) 13.0.

Array-based processing, tabular data handling, plotting, and geospatial raster and vector operations were performed using Python libraries including NumPy, pandas, Matplotlib, Rasterio, GeoPandas, and Shapely.

Model training and large-area prediction were performed on the EVE high-performance computing (HPC) cluster of the Helmholtz Centre for Environmental Research (UFZ). Computations were accelerated using an NVIDIA A100 80 GB Peripheral Component Interconnect Express (PCIe) graphics processing unit (GPU). This computational environment enabled training of the YOLO26x-seg model with 1024×1024 pixel inputs and batch sizes of 16, as well as multi-phase inference over large RGBI orthophoto datasets.

3 Results

This chapter presents the results in the order of the research questions introduced in Section 1.7. First, the effect of sequential hyperparameter tuning on mask-segmentation performance is assessed. The final post-processed prediction workflow is then evaluated across the complete evaluation set and within the Leipzig evaluation area. Next, prediction performance is compared between urban/non-forest and forest contexts, supported by selected qualitative examples. Finally, the Leipzig-wide individual tree crown map and the spatial patterns and summary statistics derived from it are presented.

3.1 Effects of hyperparameter tuning

Sequential hyperparameter tuning improved all reported mask-segmentation metrics compared with the untuned baseline model (Table 3.1).

Table 3.1 Mask-segmentation performance of the untuned baseline and tuned YOLO26x-seg models on the validation and held-out test sets.

Metric	Validation			Test		
	Untuned	Tuned	Δ	Untuned	Tuned	Δ
Precision	0.751	0.778	+0.027	0.712	0.784	+0.072
Recall	0.673	0.709	+0.036	0.683	0.695	+0.012
F1	0.710	0.742	+0.032	0.697	0.737	+0.040
mAP50	0.736	0.784	+0.048	0.723	0.773	+0.050
mAP50–95	0.423	0.483	+0.060	0.404	0.460	+0.056

On the validation set, mask precision increased from 0.751 to 0.778, corresponding to an absolute increase of 0.027, while mask recall increased from 0.673 to 0.709, corresponding to an increase of 0.036. This raised the mask F1 score from 0.710 to 0.742, an absolute improvement of 0.032. Mask mAP50 increased from 0.736 to 0.784, corresponding to an increase of 0.048, while mask mAP50–95 showed the largest absolute improvement, increasing by 0.060 from 0.423 to 0.483.

The improvements were also observed on the held-out test set. Mask precision increased from 0.712 for the untuned model to 0.784 for the tuned model, corresponding to an absolute increase of 0.072. Mask recall increased more moderately by 0.012, from 0.683 to 0.695. Consequently, the mask F1 score improved by 0.040, from 0.697 to 0.737. Test mask mAP50 increased by 0.050, from 0.723 to 0.773, while mask mAP50–95 increased by 0.056, from 0.404 to 0.460.

Validation fitness varied more strongly for some hyperparameters than for others during the individual tuning stages (Appendix Figure A.1). The initial learning rate produced one of the clearest changes in validation fitness, while an excessively high MuSGD momentum resulted in a substantial decline in performance. Visible differences were also observed for parameters such as scale augmentation, vertical flipping, mosaic augmentation, and the close-mosaic epoch. In contrast, class-positive weight, the hue, saturation, and value (HSV) augmentations, dropout, random erasing, and blue, green, and red (BGR) channel-order augmentation produced little or no measurable change in validation fitness within the investigated ranges.

The tuned model therefore outperformed the baseline across all reported validation and test metrics and was used for the subsequent prediction workflow.

3.2 Performance of the final prediction workflow

The performance of the final post-processed prediction workflow was evaluated on the entire evaluation dataset and for the Leipzig evaluation area (Table 3.2). The complete validation and test sets comprised 60 and 59 tiles, containing 3,038 and 3,517 reference crowns, respectively. The Leipzig evaluation area included 43 validation tiles with 1,664 reference crowns and 43 test tiles with 1,834 reference crowns.

Table 3.2 Performance of the final post-processed prediction workflow for the complete evaluation set and the Leipzig evaluation area.

Metric	Complete evaluation set		Leipzig evaluation area	
	Validation	Test	Validation	Test
Precision	0.70	0.67	0.70	0.67
Recall	0.65	0.66	0.67	0.66
F1	0.68	0.66	0.69	0.66
Mean matched IoU	0.77	0.75	0.77	0.77
Crown-count ratio	0.92	0.99	0.97	0.99
Crown-area ratio	0.93	0.92	0.99	0.98

Only minor differences were observed between the validation and held-out test results. For the complete evaluation set, the workflow achieved a precision of 0.67, a recall of 0.66 and an F1 score of 0.66 on the held-out test set. The mean matched IoU was 0.75. The crown-count ratio was 0.99, indicating that the total number of predicted crowns was close to the reference count. The crown-area ratio was 0.92, corresponding to an underestimation of the total reference crown area by around 8%.

The Leipzig evaluation area showed the same test precision, recall, and F1 score as the complete evaluation set. However, the mean matched IoU was slightly higher at 0.77. The crown-count ratio was 0.99 and the crown-area ratio was 0.98. Thus, the final workflow reproduced the aggregated crown count and crown area particularly closely within the area directly relevant to the Leipzig-wide map.

3.3 Performance across urban/non-forest and forest contexts

The final prediction workflow was also evaluated separately for the urban/non-forest and forest test contexts (Table 3.3). The definitions of these contexts are described in Section 2.7.4. Of the 59 test tiles, 49 belonged to the urban/non-forest context and 10 to the forest context, containing 2,081 and 1,436 reference crowns, respectively.

Table 3.3 Pooled performance of the final post-processed prediction workflow on the test set, reported separately for the urban/non-forest and forest contexts.

Metric	Urban/non-forest context	Forest context
Precision	0.64	0.71
Recall	0.69	0.63
F1	0.66	0.66
Mean matched IoU	0.77	0.73
Crown-count ratio	1.06	0.89
Crown-area ratio	0.97	0.83

The forest context achieved the higher precision, with a value of 0.71 compared with 0.64 for the urban/non-forest context. In contrast, recall was higher in the urban/non-forest context at 0.69, compared with 0.63 in the forest context. These opposing differences resulted in the same pooled F1 score of 0.66 for both contexts. The mean matched IoU was also higher in the urban/non-forest context, with values of 0.77 and 0.73, respectively.

The aggregated crown totals were reproduced more closely in the urban/non-forest context. Its crown-count ratio of 1.06 indicated a slight overprediction of the number of crowns, while its crown-area ratio of 0.97 showed close agreement with the summed reference crown area. In the forest context, the crown-count ratio of 0.89 and crown-area ratio of 0.83 indicated that both the number of individual crowns and their summed area were underestimated.

Qualitative prediction examples

The selected qualitative examples show how prediction performance varied within different spatial patterns. In the open urban example from Leipzig, most crowns were spatially separated and were individually delineated by the prediction (Figure 3.1). All reference crowns were detected, resulting in a recall of 1.00. With a precision of 0.87, the example achieved an F1 score of 0.93. Here, the predicted crown boundaries generally corresponded very closely to the reference polygons.

(a) Leipzig open urban example



Figure 3.1 Qualitative prediction example from an open urban area in Leipzig. Reference crown polygons are shown together with the corresponding post-processed predictions.

Performance was lower in the Leipzig park example, where tree crown cover was higher and neighbouring crowns were more closely connected (Figure 3.2). The example achieved a precision of 0.78, a recall of 0.77, and an F1 score of 0.78. Although many predicted crown boundaries appeared visually plausible, several predictions did not correspond one-to-one with the manually delineated reference crowns under the applied IoU-based matching procedure.

(b) Leipzig high-cover park example

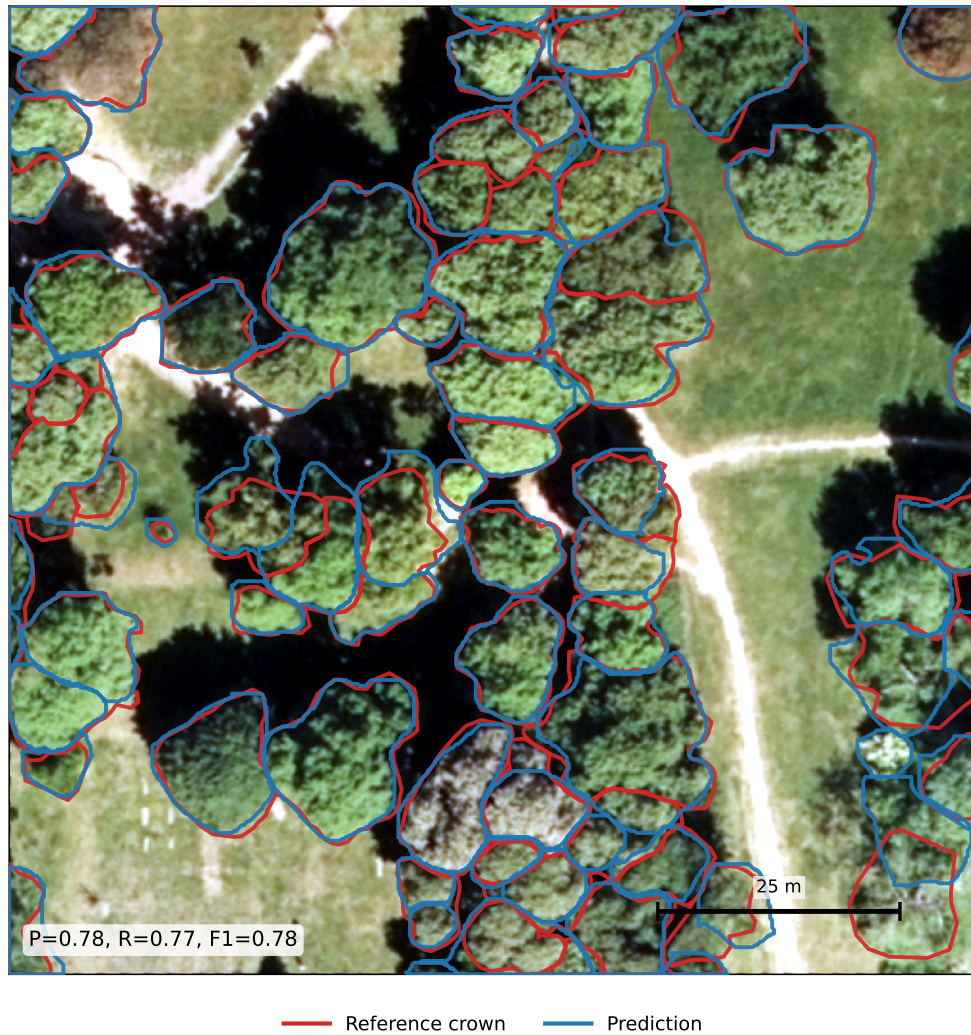


Figure 3.2 Qualitative prediction example from a park area with high tree crown cover in Leipzig. Reference crown polygons are shown together with the corresponding post-processed predictions.

The largest differences between the predictions and reference data were visible in the densely forested example from the Leipziger Auwald. Unlike Figures 3.1 and 3.2, in which reference and predicted polygons were displayed together, the Auwald reference crowns and predictions were shown separately in Figures 3.3 and 3.4 to avoid excessive visual overlap and maintain readability. The reference area contained 198 manually delineated crowns (Figure 3.3).

(c) Auwald Leipzig dense forest example: reference crowns

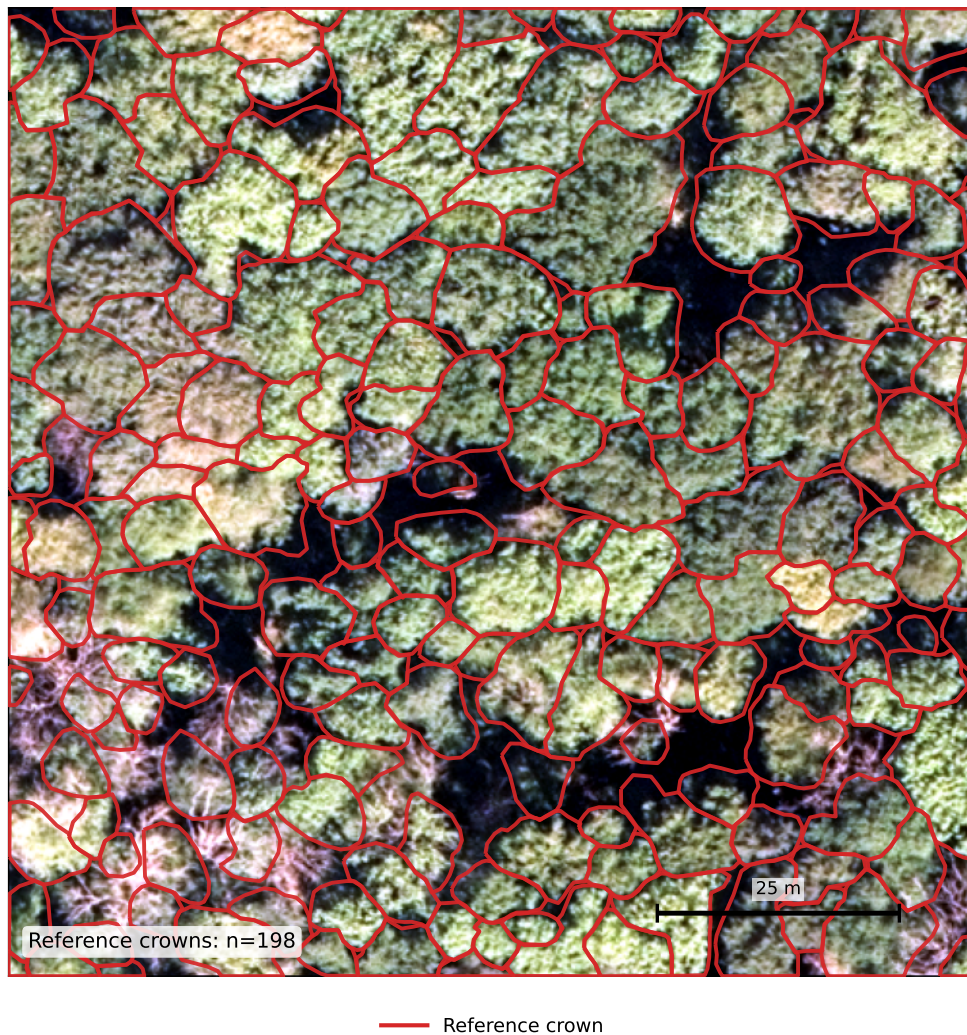


Figure 3.3 Reference tree crown polygons for a densely forested area in the Leipziger Auwald. The displayed area contained 198 manually delineated reference crowns.

The corresponding prediction achieved a precision of 0.71, a recall of 0.49, and an F1 score of 0.58 (Figure 3.4). Many reference crowns were not detected by an individual prediction, contributing to the comparatively low recall. In addition, several predicted polygons covered areas containing multiple reference crowns.

(d) Auwald Leipzig dense forest example: prediction

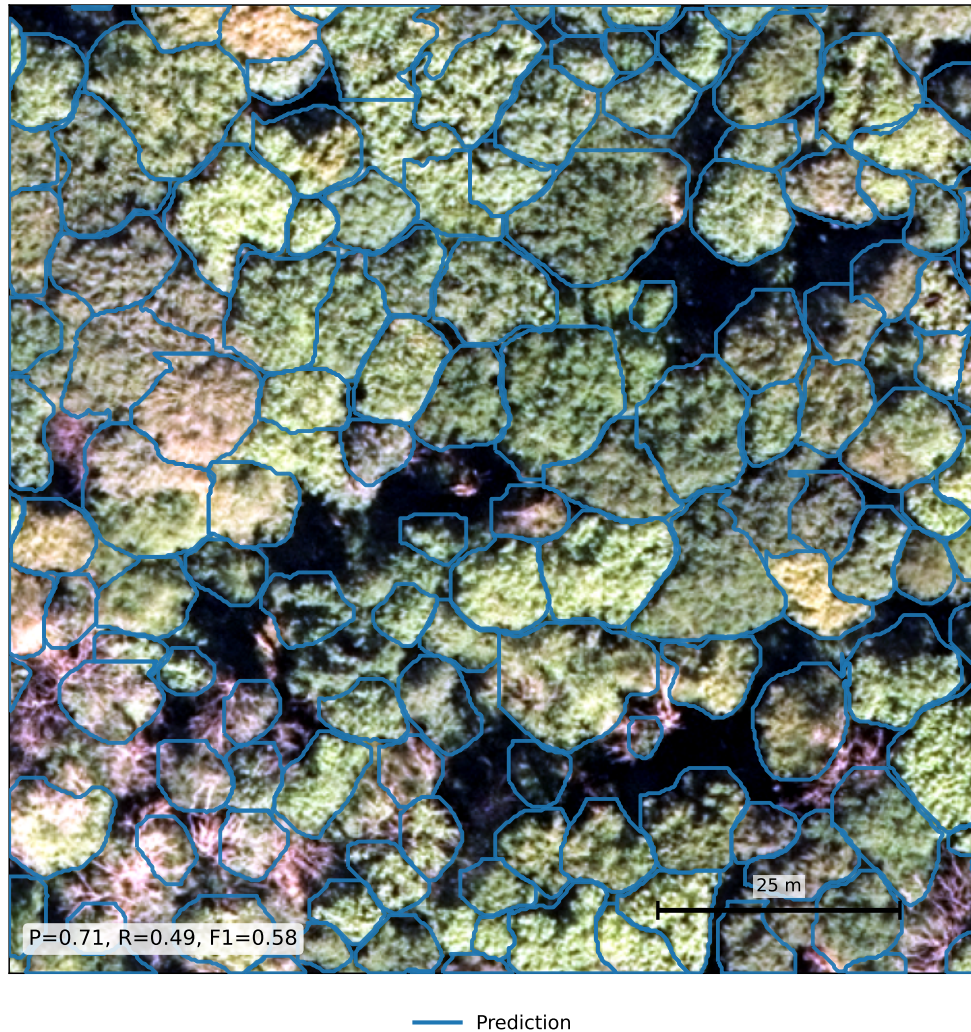


Figure 3.4 Post-processed tree crown predictions for the densely forested example in the Leipziger Auwald.

3.4 Leipzig-wide individual tree crown map

The final prediction workflow was applied to the available RGBI orthophoto coverage of Leipzig to produce a city-wide map of individual tree crowns (Figure 3.5). The largest concentrations of mapped crowns are predicted in the districts Alt-West and Süd, particularly in the Leipziger Auwald. In densely built-up and non-vegetated areas, crown cover was more fragmented or sparse.

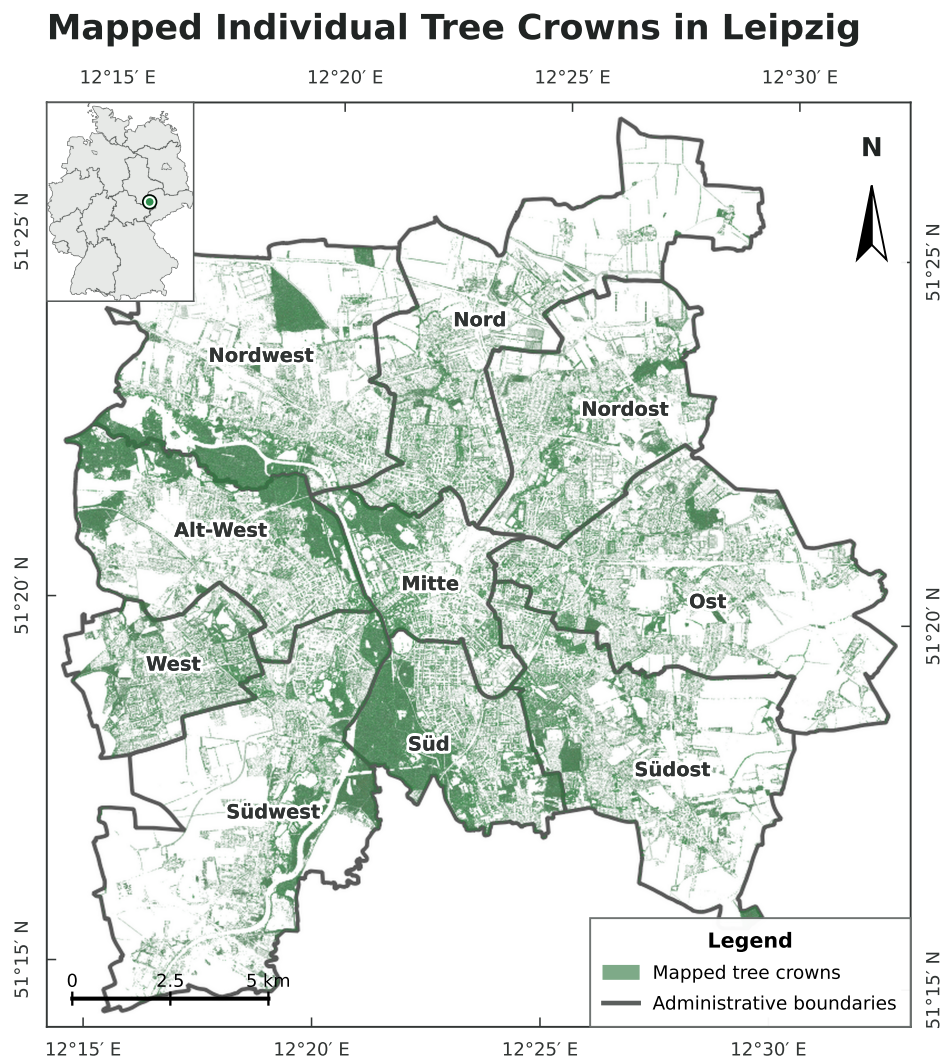


Figure 3.5 Final individual tree crown map produced for Leipzig.

The corresponding digital object identifier (DOI) is provided in the Data and code availability section.

3.4.1 Leipzig-wide crown summary and crown-size distribution

The final prediction workflow mapped a total of 951,265 individual tree crowns across Leipzig, with a summed crown area of 4,819.0 ha (Figure 3.6). The mapped crowns covered 16.2% of the Leipzig city area. The median individual crown size was 38.7 m², while the mean crown size was higher at 50.7 m².

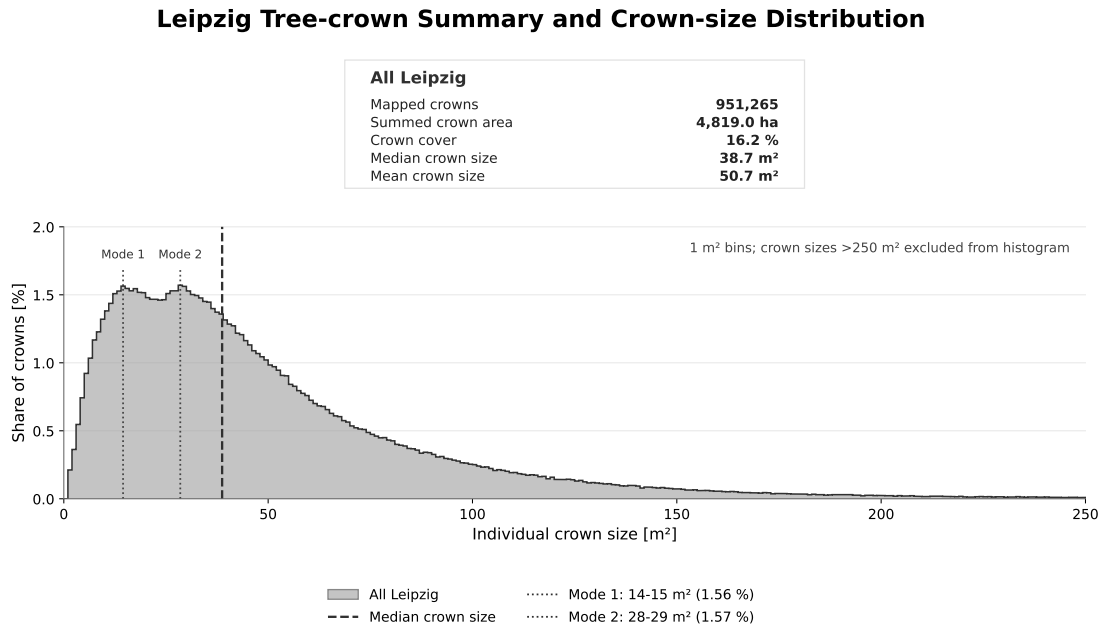


Figure 3.6 Leipzig-wide summary and distribution of mapped individual crown sizes. The histogram shows the percentage of crowns within 1 m² crown-size bins.

The distribution of mapped crown sizes was strongly right-skewed, with a large proportion of crowns occurring in the lower crown-size classes and a progressively declining share towards larger crowns. This pattern was also reflected in the mean crown size being 12.0 m² higher than the median. Two local modes were visible in the distribution, at crown sizes of 14–15 m² and 28–29 m². These bins contained 1.56% and 1.57% of all mapped crowns, respectively.

3.4.2 Crown characteristics by spatial context

For the spatial-context analysis, the mapped crowns were assigned to either forest or urban/non-forest areas, as described in Section 2.7.6 (Figure 3.7). Urban/non-forest areas contained 636,226 mapped crowns with a summed crown area of 3,114.1 ha. Forest areas contained fewer crowns in absolute terms, with 315,039 mapped crowns and a summed crown area of 1,705.0 ha.

Leipzig Tree-crown Summary and Crown-size Distribution by Spatial Context

Separated into urban / non-forest areas and forest areas

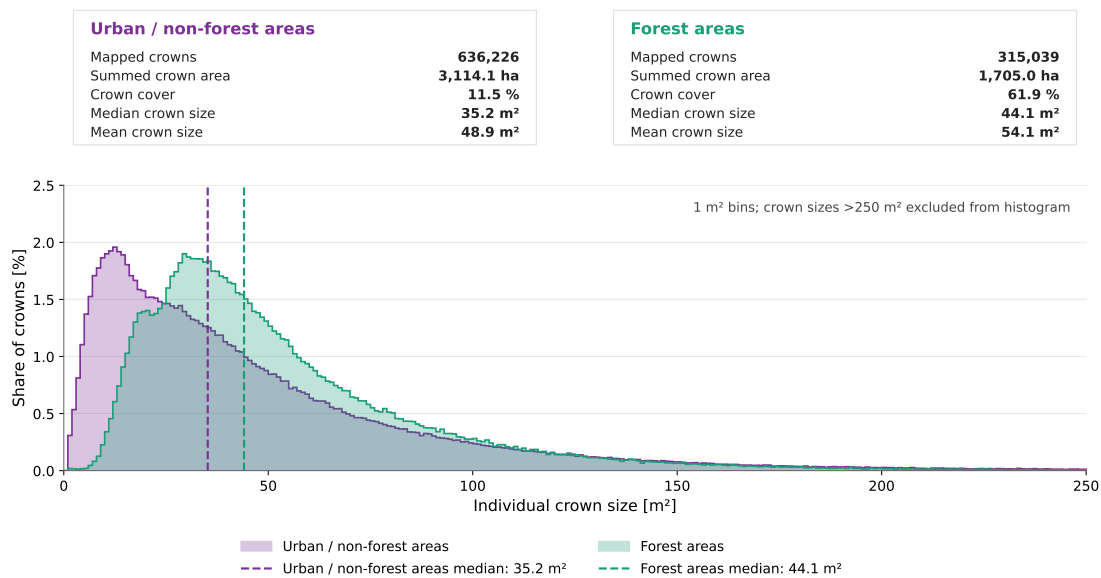


Figure 3.7 Summary and distribution of mapped individual crown sizes in forest and urban/non-forest areas of Leipzig. The histogram shows the percentage of crowns within 1 m² crown-size bins for each spatial context.

Despite the larger absolute number and summed area of crowns outside the forest areas, mapped crown cover was substantially higher within forests. Forest areas had a crown cover of 61.9%, compared with 11.5% in urban/non-forest areas. Relative to the total area of each spatial context, mapped crowns therefore occupied a much larger proportion of forest areas than of urban/non-forest areas.

Individual mapped crowns were also generally larger in forest areas. The median crown size was 44.1 m² in forest areas and 35.2 m² in urban/non-forest areas. The corresponding mean crown sizes were 54.1 and 48.9 m², respectively. Both distributions were right-skewed and overlapped considerably, but the forest distribution was shifted towards larger crown-size classes. In contrast, trees with very small crown-sizes were only predicted in urban/non-forest areas.

Each context-specific crown-size distribution showed one dominant peak and appeared unimodal. The urban/non-forest distribution reached its highest share at approximately 12–13 m², while the forest distribution peaked at approximately 29–30 m². This contrasted with the Leipzig-wide distribution in Figure 3.6, in which two local modes were visible.

4 Discussion

The following discussion interprets the results in relation to the study objectives and previous research. It first considers the improvements achieved through hyperparameter tuning and the reliability of the complete prediction workflow, before examining the different prediction behaviours observed in urban/non-forest and forest contexts. The Leipzig-wide crown totals, crown-size patterns, and spatial-context differences are then evaluated with regard to the identified sources of uncertainty. Finally, the methodological strengths and limitations of the study are assessed, followed by the implications of the resulting crown map and workflow and priorities for future research.

4.1 Model improvement through hyperparameter tuning

Sequential hyperparameter tuning improved the mask-segmentation performance of YOLO26x-seg compared with the untuned baseline across all reported validation and test metrics (Table 3.1), as presented in Section 3.1. This indicates that the selected configuration provided a more effective basis for individual tree crown segmentation. It is consistent with previous studies showing that model adaptation and fine-tuning can improve tree-crown delineation when suitable reference data are available (Burmeister et al., 2025; Liang et al., 2026; Wang et al., 2025).

The test results suggest that tuning improved the reliability of the predicted crown instances in particular. Mask precision increased more than mask recall, suggesting that the tuned model produced fewer false positives, while the proportion of correctly detected reference crowns increased only moderately. Mask mAP50–95 also improved clearly on both the validation and test sets. Because this metric includes stricter IoU thresholds, its improvement suggests that the tuned model produced masks that matched the reference crowns more closely.

The tuning results showed that some hyperparameters influenced validation fitness much more than others (Appendix Figure A.1). The clearest changes were observed for the initial learning rate and MuSGD momentum, while scale augmentation, vertical flipping, mosaic augmentation, and the close-mosaic epoch also affected

performance. In contrast, validation fitness remained unchanged across the tested values for class-positive weight, HSV hue, HSV saturation, HSV value, dropout, random erasing, and BGR augmentation. These seven parameters were therefore considered unlikely to have had a relevant influence within the investigated ranges. One possible explanation is that the task involved only a single class of tree crowns represented in high-resolution aerial orthophotos, which may have reduced the importance of class weighting and some colour-based augmentations. However, these parameters may still have an effect outside the tested ranges or in combination with other hyperparameters.

The sequential one-factor-at-a-time strategy was a practical way of investigating a large number of hyperparameters. It also made it easier to inspect the effect of each tuning stage than would have been possible in a fully combined search (Frey et al., 2003). However, the selected value of each parameter depended on the configuration inherited from the preceding tuning stages. Potential interactions between hyperparameters were therefore not investigated systematically, and a different tuning order could potentially have produced a different final configuration. Consequently, the selected configuration should be regarded as a practically optimised solution rather than a demonstrated global optimum.

The improvements reported for the tuned model describe performance before the complete large-area prediction workflow was applied. The final crown map depended not only on the trained YOLO26x-seg model, but also on multi-phase sliding-window inference, confidence-threshold selection, polygon-based IoU-NMS, and overlap subtraction. Hyperparameter tuning improved the underlying model on which the workflow was based, while the quality of the final crown polygons also depended on the subsequent inference and post-processing steps. For this reason, the complete post-processed prediction workflow was evaluated separately and is discussed in the following section.

4.2 Meaning and reliability of final workflow performance

The post-processed prediction workflow achieved an F1 score of 0.66 and a mean matched IoU of 0.75 on the complete held-out test set, as reported in Section 3.2. The similarity between the validation and test results indicates that performance remained relatively stable when the final fixed workflow was applied to the held-out data. Performance within the Leipzig evaluation area was similar to that of the complete evaluation set, with the same test F1 score and a slightly higher mean matched IoU of 0.77. The Leipzig test crown-count ratio of 0.99 and crown-

area ratio of 0.98 further show that the aggregated number and summed area of predicted crowns closely matched the reference data within the area most directly relevant to the final Leipzig map.

However, close agreement in crown count and summed crown area does not mean that every individual crown was delineated correctly. Count and area ratios describe aggregated quantities, meaning that different errors can partly compensate for one another. For example, missed crowns may be balanced by additional unmatched predictions, while merged or oversized predictions may compensate for missing or smaller crown polygons in the summed area. The object-level F1 score therefore provides important additional information, showing that many predicted and reference crowns could not be matched one-to-one. Mean matched IoU also describes only the crowns that were successfully matched and excludes unmatched predictions and reference crowns. Combining object-level and aggregated metrics consequently provides a more complete assessment of the resulting crown map than either type of metric alone.

4.3 Context-dependent prediction behaviour

The comparison between the urban/non-forest and forest test contexts showed that the same F1 score of 0.66 resulted from different error patterns, as presented in Section 3.3. The urban/non-forest context achieved higher recall but lower precision, whereas the forest context achieved higher precision but lower recall. The similar F1 scores therefore concealed important differences in the types of errors produced in the two contexts.

In the urban/non-forest context, the precision of 0.64 and crown-count ratio of 1.06 indicate that the workflow produced slightly more crowns than were present in the reference data and that a comparatively larger proportion of predictions remained unmatched.

The selected open urban example in Figure 3.1 performed better than the pooled urban/non-forest context, achieving a precision of 0.87 and detecting all reference crowns. Nevertheless, the example illustrates two reasons why an urban prediction may be counted as a false positive. At least one unmatched prediction appears to correspond to a non-tree feature and therefore represents a genuine model error. Most of the remaining unmatched predictions in the example, however, appear to delineate visible tree crowns that were absent from the reference labels. Although these predictions may have been visually plausible, they were counted as false positives because no corresponding reference crown was available for matching. This represents annotation incompleteness, or more specifically reference-label

omission, rather than necessarily an incorrect prediction. The measured precision therefore depended not only on model performance but also on the completeness of the manually created reference data. Uncertainty in the reference delineations is not unique to the present dataset. Troles et al. (2024) reported that not all individual trees could be labelled reliably where deciduous crowns overlapped in the BAMFORESTS imagery and that some over- or under-segmentation of the reference crowns had to be expected. The unmatched but visually plausible crowns in the open Leipzig example therefore reflect a broader difficulty in creating complete and unambiguous reference data for individual crown segmentation.

The park example in Figure 3.2 shows a second form of reference-related uncertainty. Although many of the predicted crown boundaries appeared visually plausible, several predictions did not correspond one-to-one with the manually delineated crowns under the applied matching rule requiring an IoU of at least 0.50. In closely connected canopies, the visible boundary between neighbouring crowns is not always unambiguous. The reference polygons represent one possible separation of these crowns, whereas the model may produce another spatially plausible separation. A prediction can therefore fail the one-to-one matching procedure even when its delineation appears reasonable in the aerial image. This helps explain why the F1 score of the park example was lower than its visual quality alone might suggest and illustrates the broader influence of reference granularity on crown-level evaluation (Troles et al., 2024).

The forest context showed the opposite error pattern. Its precision of 0.71 was higher than that of the urban/non-forest context, but its recall was lower at 0.63. Together with a crown-count ratio of 0.89 and crown-area ratio of 0.83, these results show that the workflow underestimated both the number and summed area of crowns in the pooled forest evaluation. The higher precision therefore did not indicate better overall crown delineation. Instead, it occurred alongside fewer predictions and a larger proportion of missed reference crowns.

The reference and prediction maps from the Auwald example in Figures 3.3 and 3.4 illustrate this forest error pattern. Many reference crowns did not receive an individual matching prediction, while several predicted polygons extended across areas containing multiple reference crowns. The main difficulty was therefore the separation of individual crowns within dense and connected canopy areas. Neighbouring crowns may have similar colours and textures, while overlapping canopies can obscure the visible boundaries between individual trees (Do et al., 2026; Gao et al., 2026). Because the model was trained on optical RGBI aerial orthophotos, crown separation depended on the spectral, textural, and spatial differences visible from above. The imagery did not directly provide canopy-height information that could help distinguish vertically overlapping crowns (Liang et

al., 2026). This likely contributed to the under-segmentation and missed crowns observed in the forest context.

The error pattern observed in the Auwald is consistent with previous studies of dense canopies. Do et al. (2026) identified ambiguous boundaries, severe crown overlap, occlusion, and similar canopy textures as central difficulties in dense forest and developed an explicit contour-separation and merging procedure to address them. Sun et al. (2026) also reported lower performance for densely distributed crowns. The reduced individual-crown separation in the Auwald therefore represents a recurring limitation of optical tree-crown segmentation rather than a problem specific to YOLO26x-seg.

The context comparison should nevertheless be interpreted with caution. The urban/non-forest test context contained 49 tiles, whereas the forest context contained only 10 tiles. Furthermore, the qualitative examples were selected to demonstrate characteristic prediction conditions and error patterns and are not statistically representative of all urban, park, or forest areas. The results therefore provide evidence of context-dependent prediction behaviour, but they do not constitute a balanced comparison of all possible urban and forest conditions. In addition, the two test groups represented broad site contexts defined from study-area membership (Section 2.7.4), rather than the ATKIS-based spatial contexts used for the Leipzig map analysis (Section 2.7.6).

4.4 Interpretation and reliability of the Leipzig-wide crown map

4.4.1 Reliability of crown totals and crown-size patterns

The final workflow produced a spatially continuous individual tree crown map for the available aerial-image coverage of Leipzig (Figure 3.5). A total of 951,265 crowns were mapped, with a summed crown area of 4,819.0 ha and a mapped crown cover of 16.2% (Figure 3.6). As discussed in Section 4.2, the Leipzig evaluation achieved a crown-count ratio of 0.99 and a crown-area ratio of 0.98 on the held-out test tiles, indicating that the workflow reproduced both aggregated quantities very closely. The Leipzig-wide total can therefore be regarded as a plausible estimate of the number of visible tree crowns represented in the aerial orthophotos. This estimate refers to crowns visible from above and is not equivalent to a stem-based inventory of all trees in Leipzig. Understorey trees hidden beneath the upper canopy cannot be detected. Therefore, correspondence between mapped crowns and individual trees may be less reliable in areas with dense, connected canopy cover.

A complete independent count of all trees in Leipzig was not available for comparison. The municipal tree cadastre contained 182,166 registered tree locations at the time of access, substantially fewer than the 951,265 crowns mapped in this study (Stadt Leipzig, Amt für Stadtgrün und Gewässer, 2026). However, the cadastre includes only trees in public street space and on municipal properties, and the city states that its inventory is not yet complete. The substantially higher remotely sensed crown estimate is therefore expected because it also includes visible crowns on private land and across forested areas. The two datasets represent different inventory concepts and should not be interpreted as directly competing estimates of the same tree population.

The mapped crown cover of 16.2% is in the same broad range as the approximately 19% canopy cover previously estimated for Leipzig by Strohbach and Haase (2012). That paper estimate was derived from colour-infrared orthophotos using an object-based classification approach. The difference of approximately 2.8 percentage points should not be interpreted as a temporal decline in Leipzig's tree cover, because the studies used different acquisition periods, mapping methods, canopy definitions, and reference information. In particular, the present value was calculated by summing individually predicted crown polygons, whereas the earlier study estimated overall canopy cover.

The Leipzig-wide crown-size distribution was strongly right-skewed. The median crown size of 38.7 m² was lower than the mean of 50.7 m², showing that most

mapped crowns occurred in smaller and medium-sized classes, while a smaller number of very large polygons increased the mean. These larger polygons may represent genuinely large tree crowns, but some may also result from connected crowns that were not completely separated by the model. Individual crown-size values should therefore be interpreted as mapped polygon areas rather than direct measurements of biological crown size in every case.

Two local modes were visible in the Leipzig-wide crown-size distribution, at approximately 14–15 m² and 28–29 m². The spatial-context comparison provides a plausible explanation for this pattern. The urban/non-forest distribution reached its highest share in a smaller crown-size class, at approximately 12–13 m², whereas the forest distribution peaked at approximately 29–30 m² (Figure 3.7). The combination of these differently centred distributions likely contributed to the two local modes observed for Leipzig as a whole. However, this interpretation is based on visual comparison of the histograms, and no formal test of modality or statistical mixture analysis was conducted.

4.4.2 Interpretation of spatial-context differences

The Leipzig-wide spatial-context analysis mapped 636,226 crowns in urban/non-forest areas and 315,039 crowns in forest areas (Section 3.4.2; Figure 3.7). The context-dependent test evaluation discussed in Section 4.3 is important for interpreting these totals. The urban/non-forest crown-count ratio of 1.06 indicated an overprediction of approximately 6%, whereas the forest ratio of 0.89 indicated an underprediction of approximately 11%. The corresponding crown-area ratios indicated an underestimation of approximately 3% in the urban/non-forest context and 17% in the forest context. The mapped urban/non-forest crown count may be slightly overestimated. In forest areas, both the crown count and the summed crown area are more likely to be underestimated.

As discussed for the Leipzig-wide estimate in Section 4.4.1, these context-specific totals refer to crowns visible from above rather than stem-based inventories of all trees. This distinction is particularly important for the forest areas, where understorey trees cannot be detected and dense, connected canopies make the one-to-one correspondence between mapped crowns and individual trees less reliable.

Although urban/non-forest areas contained more mapped crowns and a larger summed crown area, mapped crown cover was substantially higher in forest areas, at 61.9% compared with 11.5% (Section 3.4.2; Figure 3.7). Mapped crowns therefore occupied a much larger share of the forest extent, while the urban/non-forest context also included extensive built-up and other non-vegetated surfaces.

Mapped crowns were generally larger in forest areas. The median crown size was 44.1 m² in forests and 35.2 m² in urban/non-forest areas, while the corresponding mean values were 54.1 and 48.9 m². The forest distribution was therefore shifted towards larger crown-size classes. This may partly reflect differences in vegetation structure and the more continuous canopy within forest areas. However, as discussed in Section 4.3, incomplete crown separation in dense forest may have increased the apparent size of some predicted polygons. The observed difference in mapped crown size cannot consequently be attributed entirely to biological differences between forest and urban trees.

Very small mapped crowns occurred primarily in the urban/non-forest context. These polygons may represent young or ornamental trees, partly visible crowns, or small isolated crowns whose boundaries were easier to distinguish against surrounding non-vegetated surfaces. Some may also represent fragments or uncertain predictions.

Overall, the spatial-context analysis showed that the Leipzig-wide totals concealed substantial differences in crown concentration, mapped crown size, and prediction behaviour. Forest areas had much higher crown cover and generally larger mapped crowns, but their crown count and summed crown area were more likely to be underestimated. In contrast, the urban/non-forest crown count may be slightly overestimated, although its summed crown area remained close to the reference value.

4.5 Methodological strengths and limitations

4.5.1 Methodological strengths

A major strength of this study is that model development, large-area prediction, post-processing, and evaluation were treated as parts of one integrated workflow. The final evaluation was not limited to predictions on isolated input tiles. Instead, the validation and test predictions were generated using the same multi-phase sliding-window inference and polygon post-processing procedures that were applied during production of the Leipzig-wide map (Section 2.7.4). This allowed effects introduced by overlapping windows, edge filtering, duplicate removal, and overlap subtraction to be included in the final performance assessment.

The use of separate training, validation, and held-out test subsets further supported an independent assessment of the final fixed workflow. Validation data were used for hyperparameter tuning, checkpoint selection, confidence-threshold selection, and post-processing calibration, whereas the test subset was retained for final

evaluation. The similarity between validation and test performance suggests that the selected configuration did not lead to a substantial loss of performance on the held-out tiles.

The reference dataset also covered several urban and forest sites rather than a single local environment. This introduced variation in crown density, canopy structure, image acquisition, and background conditions during model development. Although Leipzig provided the largest share of the reference data, the inclusion of additional sites reduced the dependence of the model on one specific urban setting.

Another strength was the use of complementary evaluation metrics. Precision, recall, F1, and mean matched IoU described object-level detection and geometric agreement, while crown-count and crown-area ratios assessed the aggregated properties most relevant to the final map. As shown in Section 4.2, these metric groups provided different information and prevented close agreement in total crown number or area from being interpreted as evidence that every individual crown had been delineated correctly.

Finally, the workflow produced georeferenced individual crown polygons rather than only image-level predictions or a continuous canopy-cover raster. This enabled the calculation of Leipzig-wide crown counts, crown sizes, crown cover, and spatial-context statistics and provides a basis for linking the mapped crowns to further spatial or administrative information.

4.5.2 Limitations and sources of uncertainty

Despite the separation of training, validation, and test tiles, all subsets were drawn from the same broader collection of study sites. The held-out test results therefore demonstrate performance on unseen tiles within the represented sites, but not full geographic transfer to an entirely independent city, sensor, or image-acquisition campaign. Nearby tiles may also share similar vegetation structures and imaging conditions. An evaluation in a geographically independent city would provide a stronger assessment of transferability.

The manually delineated crown polygons were another important source of uncertainty. As discussed in Section 4.3, individual crown boundaries were not always unambiguous, particularly where neighbouring crowns touched or overlapped, and some visible crowns were absent from the reference data. The calculated metrics therefore measure agreement with the available reference interpretation rather than with an entirely objective ground truth (Troles et al., 2024).

The model relied exclusively on optical aerial imagery and therefore represented only crowns visible from above. Understorey trees hidden beneath the upper canopy

could not be detected, while the absence of direct canopy-height information limited the separation of vertically overlapping crowns (Liang et al., 2026). As discussed in Section 4.3, this limitation was most apparent in dense forest.

The sequential one-factor-at-a-time tuning strategy also had limitations. Although it allowed many hyperparameters to be evaluated within the available computational budget, it did not systematically assess interactions between parameters. The final configuration depended on the order of the tuning stages and should therefore be interpreted as a practically optimised solution rather than a demonstrated global optimum, as discussed in Section 4.1.

Similarly, the confidence threshold and post-processing settings were selected using validation performance and visual examination of the resulting polygons. This produced one fixed configuration that could be applied consistently throughout evaluation and map production. However, the urban/non-forest and forest results showed that the same settings produced different balances between precision and recall across contexts. Separate context-specific settings might improve local performance, but they would also make the workflow less uniform and could introduce discontinuities in a city-wide map.

The context-based evaluations and Leipzig-wide spatial analysis should also be distinguished. The evaluation contexts were defined by grouping study sites (Section 2.7.4), whereas the Leipzig-wide forest and urban/non-forest classes were derived from ATKIS forest polygons (Section 2.7.6). In addition, the forest test subset contained considerably fewer tiles than the urban/non-forest subset. The evaluation ratios therefore indicate likely directions of context-dependent bias but should not be treated as direct correction factors for the Leipzig-wide totals.

Finally, the accuracy of the complete Leipzig-wide map was inferred from the held-out reference tiles rather than assessed through spatially continuous reference data across the entire city. Local errors may therefore remain in areas or image conditions that were not represented by the evaluation tiles. The resulting dataset should be interpreted as a remotely sensed map of visible crown objects and not as a field-verified inventory of every individual tree.

4.6 Implications and future research

The main contribution of this study is not limited to the resulting Leipzig-wide crown map. The developed workflow provides an integrated approach for adapting an instance-segmentation model to four-band RGBI aerial orthophotos, applying it across large image extents, resolving predictions from overlapping inference windows, and evaluating the resulting georeferenced crown polygons. The combination of multi-phase sliding-window inference, confidence-threshold selection, polygon-based duplicate removal, and overlap subtraction provides a framework that could be adapted for individual tree crown mapping in other cities or regions. However, its transferability would need to be assessed using imagery from an independent geographic area and acquisition campaign.

The resulting crown map provides information beyond that of a continuous canopy-cover raster because its georeferenced polygons allow the number, size, area, and spatial distribution of mapped crowns to be analysed individually. The dataset can therefore support comparisons between districts or spatial contexts and be linked to information on land use, surface temperature, imperviousness, population, or socioeconomic conditions. Such analyses could help investigate the distribution of urban tree cover, heat exposure, and access to green infrastructure across Leipzig.

The crown map may also complement the municipal tree cadastre. While the cadastre contains detailed information on registered trees in public and municipally managed areas, the remotely sensed map also represents visible crowns on private property and within forested areas. Linking the two datasets could help identify differences in spatial coverage and provide additional information for matched trees. However, crown polygons and stem-based records represent different objects and would require careful spatial matching. The map should therefore not be treated as a replacement for a field-verified inventory; its main value lies at neighbourhood, district, and city scale, where aggregated crown patterns are more important than the exact correspondence between every polygon and an individual tree.

A further extension of the workflow would be to combine crown delineation with the analysis of crown-level attributes. The mapped polygons could serve as spatial units for extracting spectral, textural, and structural information from the RGBI imagery. These attributes could support tree-species classification using a separate model, or species classification and crown segmentation could be learned jointly within a multi-class instance-segmentation framework (Sun et al., 2026). Crown-level vegetation indices could additionally support the screening of comparatively low vegetation activity or possible signs of stress (Dersch et al., 2023; Khan et al., 2025; Yoon et al., 2026). Such extensions would require representative reference datasets containing reliable species or condition labels.

Repeated application of the workflow to aerial orthophotos from different years could enable crown-level change analysis. Spatially matching predictions between acquisitions could support the identification of crown loss, newly visible crowns, crown expansion, or major reductions associated with pruning, damage, or mortality. However, differences in phenology, illumination, image quality, and segmentation behaviour could also produce apparent changes. Multi-temporal applications would therefore require consistent imagery and a robust method for matching crown objects across acquisition dates.

Future methodological development should particularly address crown separation in dense and connected canopies. Additional height information from airborne laser scanning, digital surface models, or photogrammetric point clouds could help distinguish vertically overlapping crowns that cannot be separated reliably from RGBI imagery alone (Liang et al., 2026). Methods that explicitly improve the separation of touching instances could also be investigated (Do et al., 2026; Gao et al., 2026). Further improvements could include uncertainty-based selection of new training samples, additional forest reference areas, and the evaluation of adaptive or context-specific post-processing settings.

Finally, the workflow should be tested on imagery from an independent city or region. The present test set demonstrated performance on unseen tiles from the represented study sites, but not transfer to a completely new geographic domain. Crown appearance and segmentation performance may vary with tree species, canopy structure, image characteristics, acquisition season, and annotation style (Do et al., 2026; Zhao et al., 2023). The openly shared crown map, workflow, and associated code provide a basis for reproducing the present study and evaluating how well the approach can be adapted to other large-area crown-mapping applications.

5 Conclusion

This study developed and evaluated an integrated YOLO26x-Seg workflow for mapping individual tree crowns across Leipzig using 20 cm RGBI aerial orthophotos. This workflow covered the entire mapping process: data preparation; model training; sequential hyperparameter tuning; multi-phase sliding-window inference; confidence-threshold selection; polygon post-processing; evaluation of final crown polygons; and production of a Leipzig-wide map of individual tree crowns.

Sequential hyperparameter tuning improved the mask-segmentation performance of YOLO26x-seg compared with the untuned baseline. Improvements were observed across all reported validation and test metrics. On the held-out test set, mask F1 increased from 0.697 to 0.737, mask mAP50 from 0.723 to 0.773, and mask mAP50–95 from 0.404 to 0.460. The stronger increase in precision than recall indicates that tuning primarily reduced false-positive predictions, while the improvement in mAP50–95 also suggests more accurate crown-mask delineation. The selected configuration should nevertheless be understood as a practically optimised solution because the sequential one-factor-at-a-time procedure did not systematically investigate interactions between hyperparameters.

Following large-area inference and polygon post-processing, the final complete workflow achieved an F1 score of 0.66 and a mean matched IoU of 0.75 on the held-out test set. Within the Leipzig evaluation area, the workflow produced the same F1 score, as well as a slightly higher mean matched IoU of 0.77. The Leipzig crown-count ratio of 0.99 and crown-area ratio of 0.98 indicate that the aggregated number and summed area of predicted crowns closely matched the reference data. However, this close aggregate agreement did not imply correct one-to-one delineation of every crown, because missed, additional, merged, and differently separated crown polygons could partly compensate for one another.

Prediction behaviour differed substantially between spatial contexts despite the identical F1 score of 0.66 obtained for the urban/non-forest and forest test groups. The urban/non-forest context showed higher recall but lower precision and a crown-count ratio of 1.06, indicating slight overprediction of crown numbers. In contrast, the forest context showed higher precision but lower recall, with a crown-count ratio of 0.89 and crown-area ratio of 0.83. Dense and connected forest canopies therefore remained the most difficult conditions, particularly because neighbouring crowns

were not always separated into individual polygons. These findings show that a single overall performance value can conceal spatially different error patterns.

The final Leipzig-wide map contained 951,265 mapped crowns with a summed crown area of 4,819.0 ha, corresponding to 16.2% of the mapped municipal area. The median mapped crown size was 38.7 m², while the mean was 50.7 m². Urban/non-forest areas contained 636,226 mapped crowns, whereas forest areas contained 315,039. Although the absolute number of mapped crowns was higher outside forests, crown cover was substantially greater within forest areas, at 61.9% compared with 11.5%. Forest crowns were also generally represented by larger polygons. Some of this difference may reflect vegetation structure, but incomplete separation of connected forest crowns may also have increased the size of individual mapped polygons.

The resulting map represents crowns visible from above rather than a complete stem-based inventory of all trees in Leipzig. Understorey trees cannot be detected from the aerial orthophotos, and the correspondence between mapped crowns and individual trees is less reliable in dense canopy areas. Nevertheless, the close Leipzig crown-count and crown-area ratios indicate that the map provides a plausible representation of aggregated visible crown patterns. Its georeferenced polygon structure enables analyses of crown number, size, area, cover, and spatial distribution that extend beyond the information available from a continuous canopy-cover raster.

Overall, the study demonstrates that YOLO26 instance segmentation can be integrated with large-area inference, polygon post-processing, and map-level evaluation to produce a detailed individual tree crown map for a heterogeneous urban area. The main contribution is therefore both the Leipzig-wide crown dataset and the reproducible workflow used to generate it. Future research should particularly investigate improved crown separation using height or structural information, expansion of the forest reference data, crown-level species and condition analysis, multi-temporal mapping, and evaluation of the workflow in an independent city or image domain.

Data and code availability

Data availability. The final Leipzig-wide individual tree-crown dataset is published on Zenodo as *Leipzig Tree-Crown Map* (version 1) and is available at [10.5281/zenodo.21709365](https://doi.org/10.5281/zenodo.21709365). The deposit contains the final post-processed tree-crown polygons in GeoPackage format, the orthophoto GeoTIFF files included in the public release, and accompanying documentation.

Code availability. The code used for data preparation, model training, large-area inference, post-processing, evaluation, and result generation is archived on Zenodo at [10.5281/zenodo.21788398](https://doi.org/10.5281/zenodo.21788398). The corresponding GitHub repository is available at github.com/mxkopf/leipzig-tree-crown-mapping-yolo26.

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Appendix

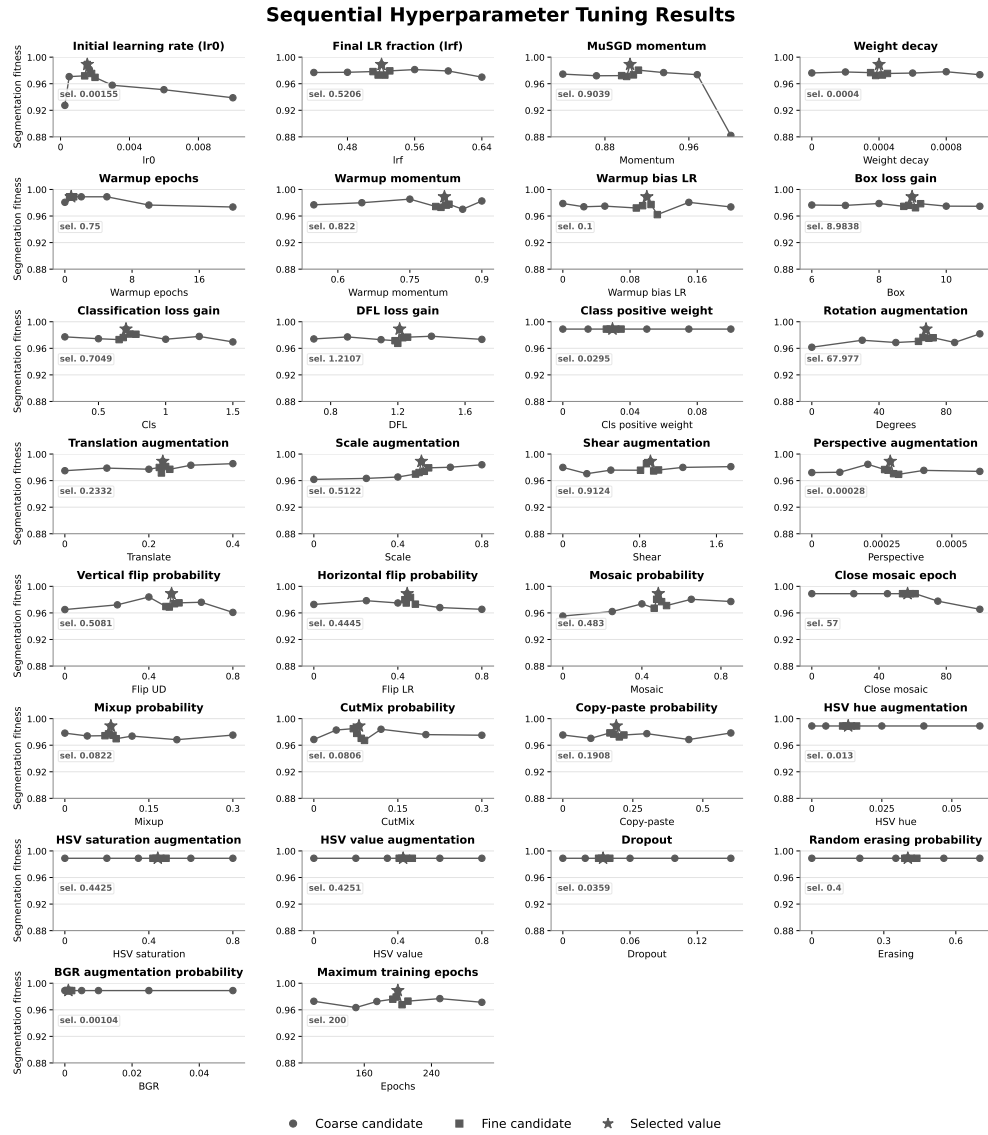


Figure A.1 Validation-fitness patterns obtained during sequential hyperparameter tuning. Each panel shows the evaluated coarse and fine candidate values and the selected value for one hyperparameter.

Statement on the use of generative artificial intelligence

I declare that generative artificial intelligence (AI) tools were used during the preparation of this master thesis only as supporting tools and not as a replacement for my own scientific work. The use of AI was limited to tasks that supported the development, organisation, and presentation of the thesis. All final decisions regarding the research design, data processing, analysis, interpretation of results, and written content were made by me.

AI tools were used to support the following tasks:

- developing, debugging, and improving code for data preparation, model training, prediction, evaluation, and visualisation workflows;
- identifying and resolving technical problems in scripts, terminal outputs, and LaTeX documents;
- improving the clarity, structure, and wording of thesis text;
- summarising and comparing scientific literature as preliminary support for writing and contextualisation;
- checking consistency between methodological descriptions, figures, tables, and references.

All AI-generated suggestions were critically reviewed, verified, and, where necessary, revised before being included in the thesis. AI outputs were treated as preliminary assistance and were not used as independent evidence. Scientific claims, methodological descriptions, results, and conclusions were checked against the underlying data, code outputs, literature, or my own interpretation.

No confidential, personal, or unpublished third-party data were intentionally provided to AI systems during the preparation of this thesis. I take full responsibility for the accuracy, integrity and analyses presented in this thesis.

Declaration

"I certify that I have written this thesis independently and only with the use of the sources and aids indicated; in particular, literal or analogous quotations are labeled as such. I am aware that non-compliance can also lead to the subsequent cancellation of the degree."

Leipzig, 04.08.2026

Place & Date

Signature